

RoboCup Standard Platform League (NAO) Rule Book

RoboCup China Technical Committee

(DRAFT 2025 rules, as of 2025-01-27)

RoboCup标准平台联赛（NAO）规则手册

RoboCup中国技术委员会

(DRAFT 2025规则，截至2025-01-27)

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1 Setup of the Environment

1.1 Field Construction

The standard soccer field consists of 8 mm artificial turf mounted on a flat wooden base with a total area of length 10.4 m and width 7.4 m. Care should be taken to ensure the field is as flat and level as possible. Additionally, the wooden base should be well-supported and should not give when humans stand or walk on it.

Figure 1 shows the dimensions of a standard soccer field. A more detailed technical drawing is provided in Appendix C to this document. Note that the penalty mark is a cross and there is a dash at the midpoint of the halfway line. White field lines can be made of the same 8 mm artificial turf, but in white (i. e., made of white artificial turf), spray-painted or taped. Regardless of the solution, the field lines must be durable throughout the competition.

Figure 2 and Figure 3 depict the construction and placement of the goals. A support structure for the net shall be made with small black, white, or gray bars or cylinders. The support structure shall be constructed exactly as shown in Figure 3.

1.2 Field Colors

The colors of the soccer field are as follows:

- The field (artificial turf) itself is green (color is not specified, but it should not be too dark).
- The lines on the field are white, whether they are taped, spray painted or made from white artificial turf.
- The posts and top crossbar of both goals are white. The net and the support structure for the net are white, gray, or black.

1.3 Lighting Conditions

The lighting conditions depend on the actual venue. SPL fields should be placed near or under windows where possible. Whether window lighting is used or not, ceiling lights should be provided as necessary to ensure that most of the field is never darker than 300 Lux (400 Lux preferred).

Natural and non-natural light must be free to reach the field. The technical committee can delimit a zone near the field where humans must not stand and where any items blocking the light sources are forbidden.

1 环境设置

1.1 场地建设

标准足球场由8毫米的人工草坪铺设在平坦的木质基础上，总面积为10.4米长和7.4米宽。应确保场地尽可能平坦。此外，木质基础应得到良好支撑，人在其上站立或行走时不应变形。

图1显示了标准足球场的尺寸。本文件的附录C提供了更详细的技术图纸。请注意，罚球点是十字形，中线中点处有一条短线。白色场地线可由相同的8毫米人工草坪制成，但需为白色（即使用白色人工草坪），或喷漆或贴带。无论采用何种方案，场地线都必须在整个比赛中保持耐用。

图2和图3描绘了球门的建设和安装。网状结构的支撑应使用黑色、白色或灰色的细杆或圆柱。支撑结构必须严格按照图3所示进行建造。

1.2 场地颜色

足球场的颜色如下：

- 场地（人造草坪）本身是绿色的（颜色未指定，但不应过暗）。
- 场地上的线条是白色的，无论是贴的、喷绘的还是用人造草坪做的人造草坪。
- 两个球门的立柱和顶横梁都是白色的。球网和支持球网的支架是白色的、灰色的或黑色的。

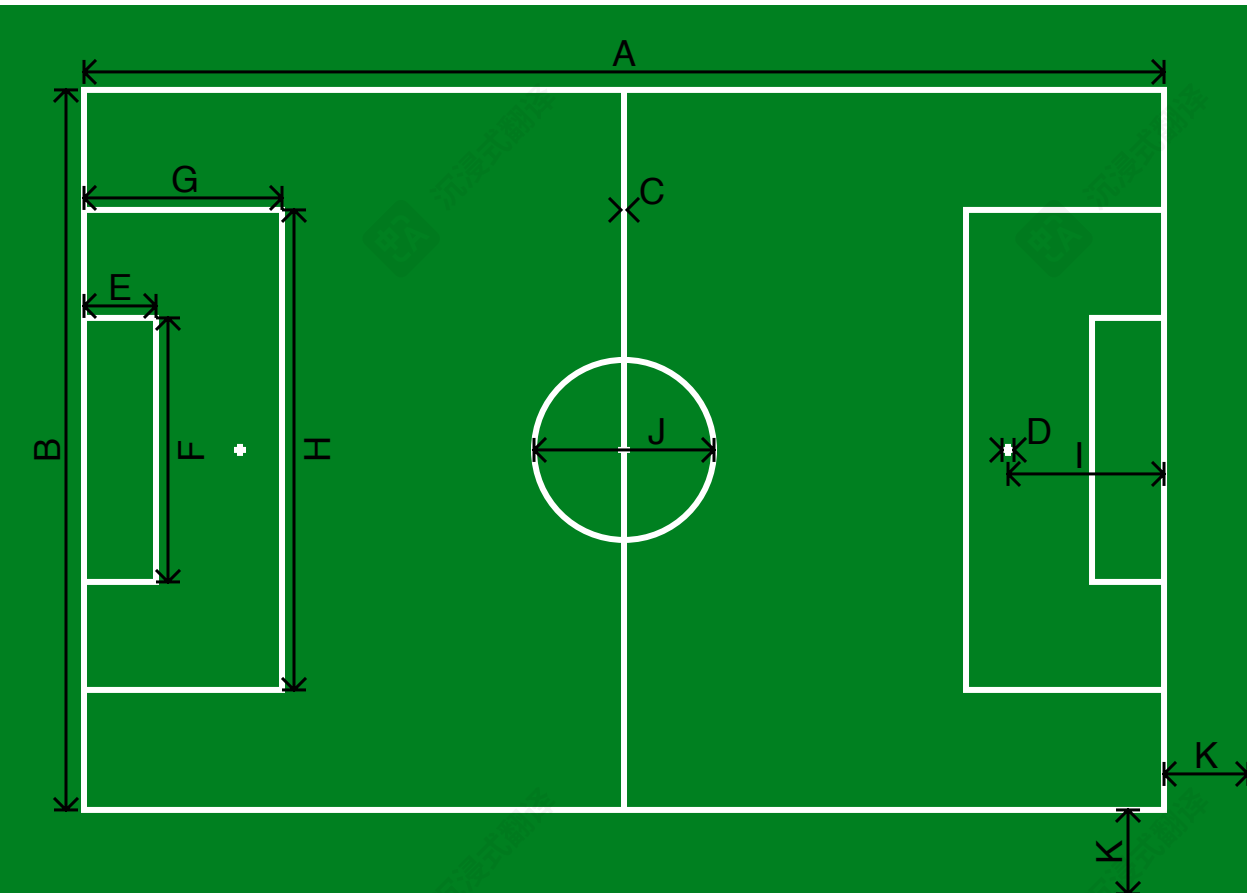
1.3 光照条件

光照条件取决于实际场地。SPL场地应尽可能放置在靠近或窗户下方。无论是否使用窗户光照，都应根据需要提供天花板灯，以确保场地大部分区域的光照不低于300勒克斯（推荐400勒克斯）。

自然光和非自然光必须能够自由照射到场地。技术委员会可以划定一个靠近场地的区域，禁止人类在此站立，并且禁止任何阻挡光源的物品。

1.4 Venue Setup

Fields may be located close to one another. Barriers will not necessarily be constructed between adjacent fields to block the robots from seeing other fields, goals, or balls. However, barriers will be constructed to block sight between any fields that are not located at least three meters apart. Hence, for each side of a field that is adjacent to another field, either barriers will separate the fields or at least 3 m will be between the carpet of adjacent fields.

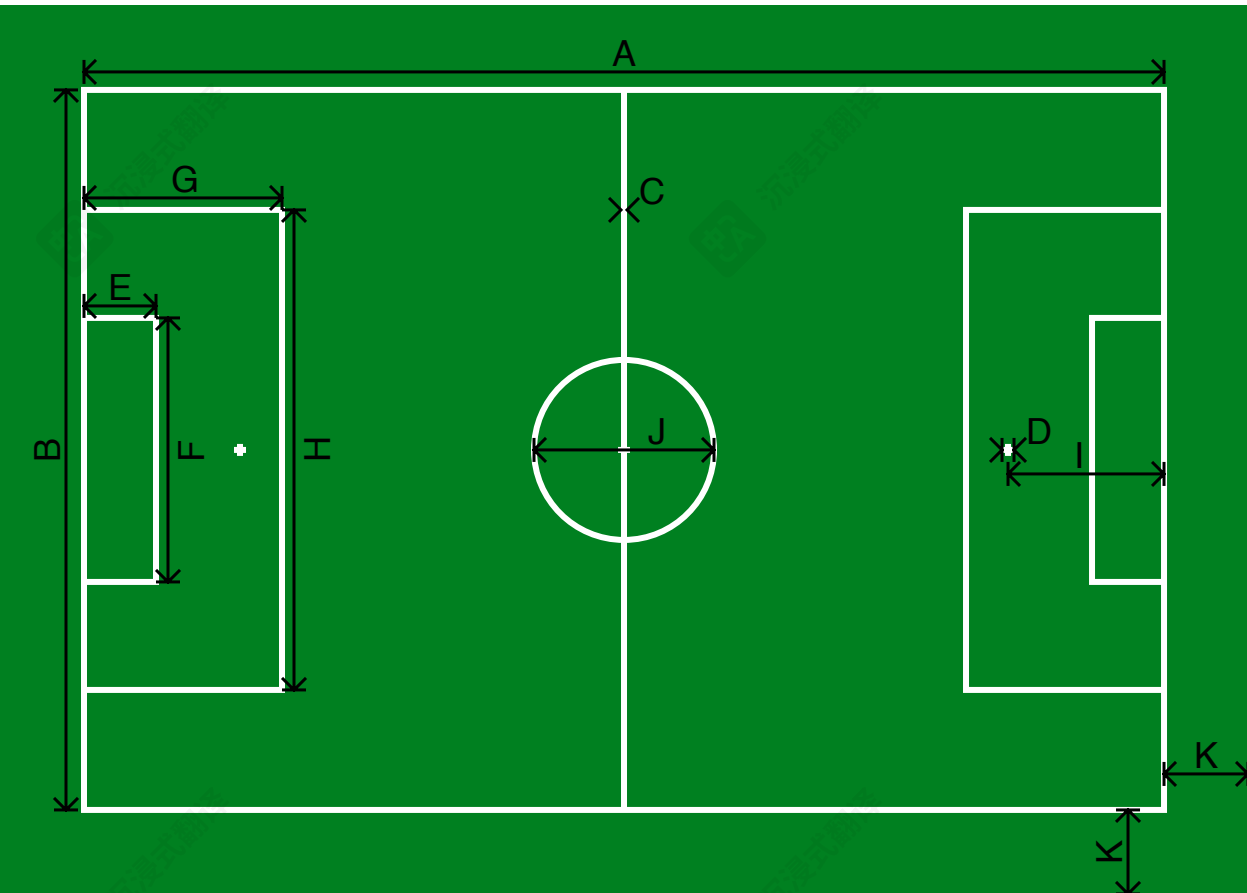


ID	Description	Length (in mm)	ID	Description	Length (in mm)
A	Field length	9000	G	Penalty area length	1650
B	Field width	6000	H	Penalty area width	4000
C	Line width	50	I	Penalty mark distance	1300
D	Penalty mark size	100	J	Center circle diameter	1500
E	Goal area length	600	K	Border strip width	700
F	Goal area width	2200			

Figure 1: Schematic diagram of the soccer field (not to scale) and corresponding dimensions in mm. Note that measurements on this diagram are made to the center of lines.

1.4 场地布置

场地可以彼此靠近。相邻场地之间不一定需要建造障碍物来阻挡机器人看到其他场地、目标或球。然而，对于至少相距三米以上的场地，将建造障碍物来阻挡视线。因此，对于与另一个场地相邻的场地的每一边，要么障碍物将分离两个场地，要么相邻场地的地毯之间至少有3米的距离。



ID	描述	长度（毫米）	ID	描述	长度（毫米）
A	场地长度	9000	G	罚球区长度	1650
B	场地宽度	6000	H	罚球区宽度	4000
C	线宽	50	I	罚球点距离	1300
D	罚球点大小	100	J	中圈直径	1500
E	球门区长度	600	K	边带宽度	700
F	球门区宽度	2200			

图1：足球场示意图（未按比例）及对应毫米尺寸。注意该图中的测量以线为中心。

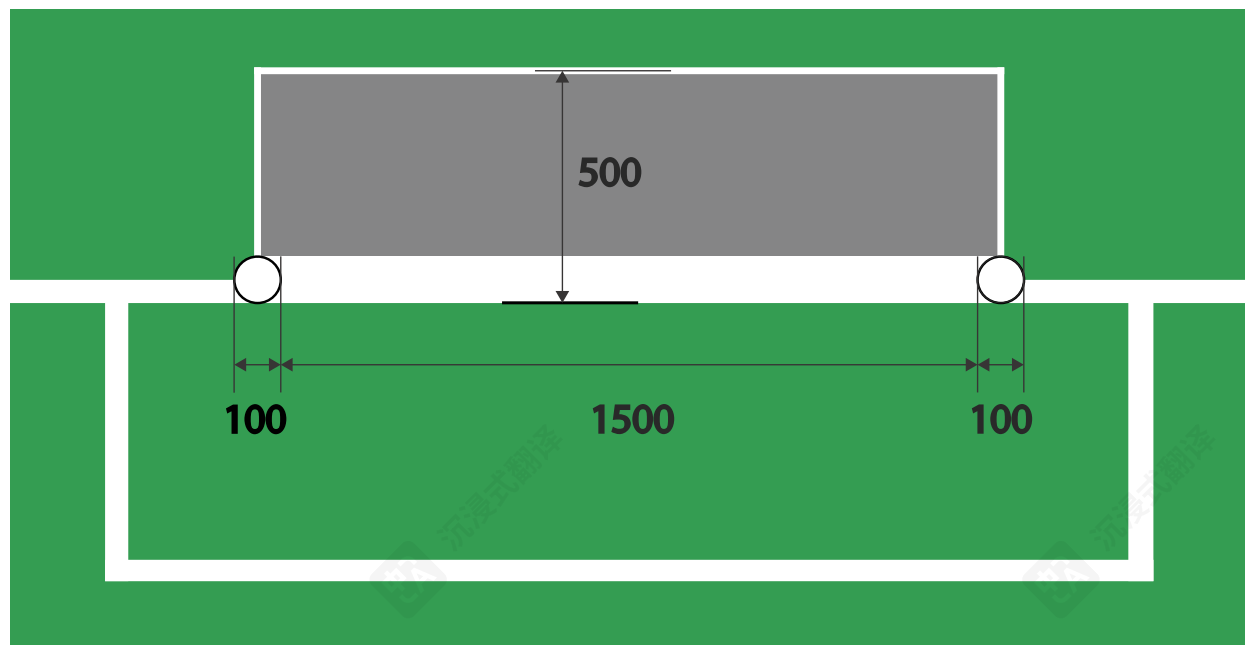


Figure 2: Dimensions of the goal (in mm), viewed from above, and its placement on the field.

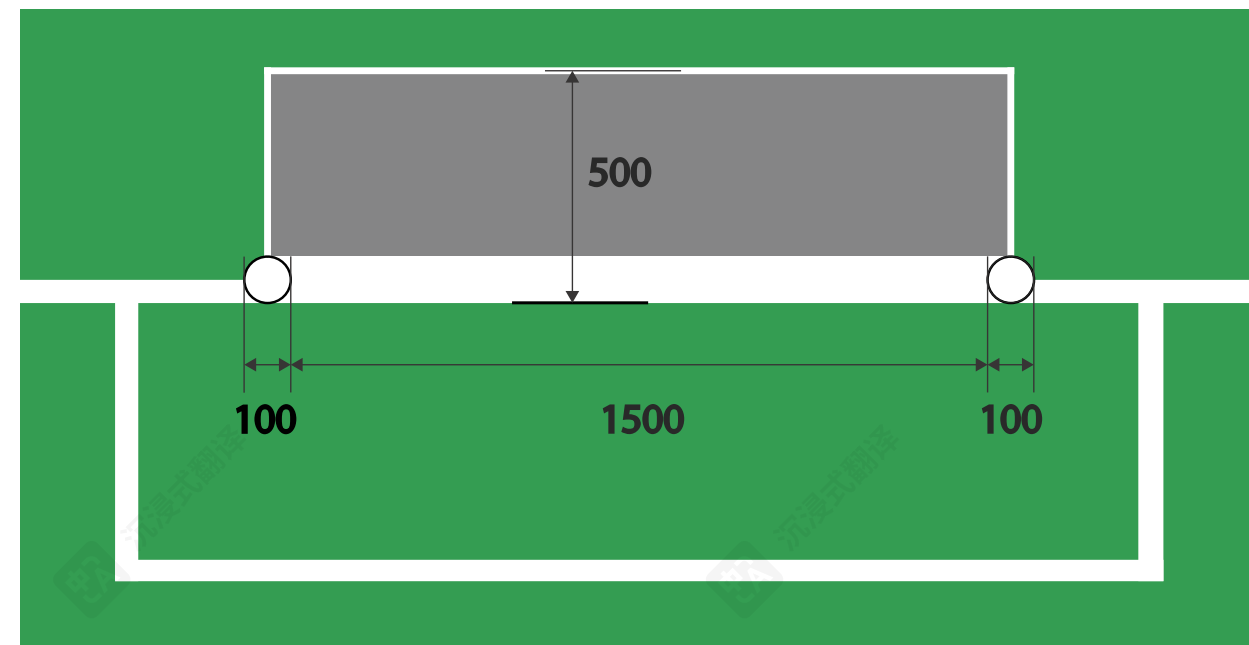
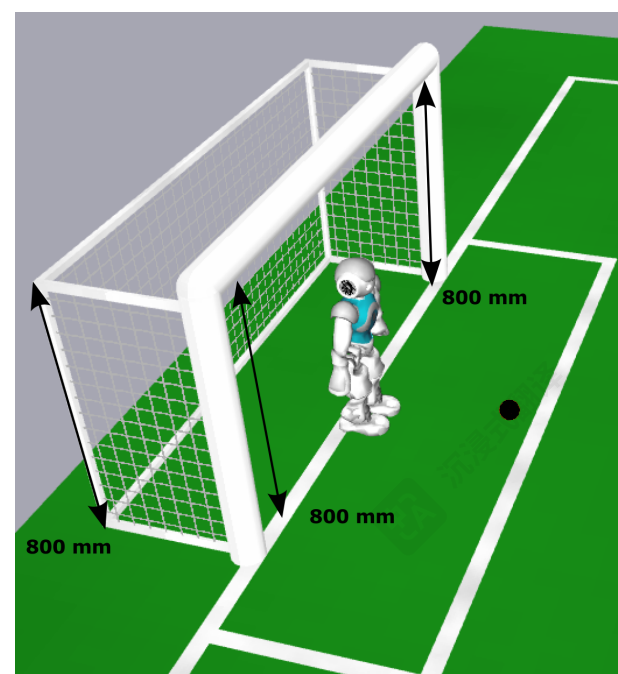


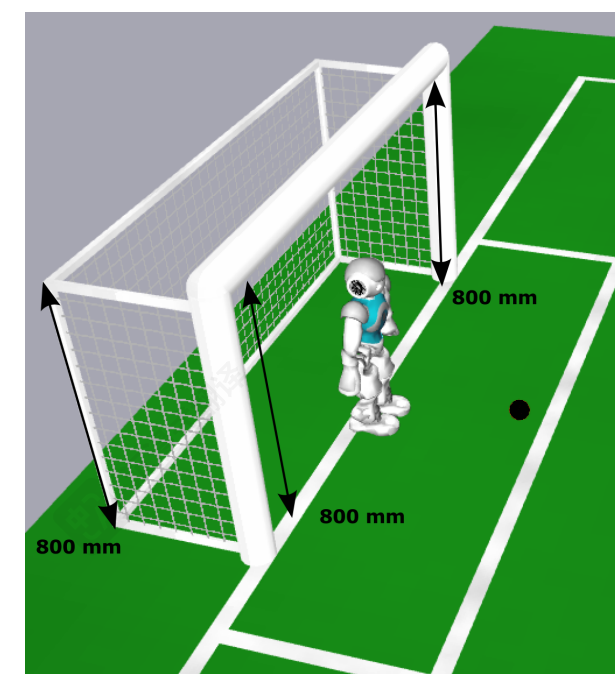
图2：球门尺寸（单位：毫米），从上方视角查看，及其在场上的位置。



The goalposts and crossbar are made from 3 white cylinders with a diameter of 100 mm. The net:

- has a height of 800 mm
- is of white, gray or black color
- is tightly supported via the support structure, in a way to minimize interference with the goalkeeper
- has a weave with holes smaller than the ball diameter.

Figure 3: Appearance and dimensions of the goals.



球门柱和横梁由3个直径为100毫米的白色圆柱制成。网：

- 的高度为800毫米
- 是白色、灰色或黑色
- 通过支撑结构紧密支撑，以最大限度减少对守门员的干扰
- 的网眼孔径小于球体直径。

图3：球门的外观和尺寸。



Figure 4: A NAO and an official ball.



图4：一个NAO和一个官方球。

1.5 Ball

The official ball is a soft foam ball with a black and white soccer ball print (see Figure 4). They are 100 mm in diameter and weigh 44 g. These balls are available by writing to info@sportpaint.de (in German or English) and asking to order the “pu schaumstoffball 10cm 100ss”. Each ball costs €2.86 plus shipping, where shipping cost depends on the destination.

1.6 Definition of Inside and Outside

A line is always part of a region of the field. This means, that *inside/outside* <region> refers to the green area as well as the surrounding line. Specifically:

- The goal lines and touchlines are part of the field.
- The goal area lines (and the adjacent part of the goal line) are part of the goal area.
- The penalty area lines (and the adjacent part of the goal line) are part of the penalty area.
- The center circle lines are part of the center circle.

Notable exceptions to this rule are:

- The halfway line does not form part of any half. That is, a robot is *outside* a half of the field if it is touching the halfway line.
- The goal line is not part of the goal.

1.5 球

官方用球是一个柔软的泡沫球，带有黑白足球图案（见图4）。它们直径为100毫米，重量为44克。这些球可以通过发送邮件至info@sportpaint.de（德语或英语）并要求订购“pu schaumstoffball 10cm 100ss”来获得。每个球的价格为 €2.86元，加上运费，运费取决于目的地。

1.6 内部和外部的定义

一条线始终是场地区域的一部分。这意味着，*inside/outside* <region> 区域指的是绿色区域以及周围的线条。具体来说：

- 球门线和边线是场地的一部分。
- 球门区线条（以及相邻的球门线部分）是球门区的一部分。
- 罚球区线（以及球门线相邻部分）属于罚球区。
- 中圈线属于中圈。

这条规则有显著例外：

- 中线不属于任何半场。也就是说，如果一个机器人触碰到中线，则它处于场地的半场之外。
- 球门线不属于球门的一部分。

1.7 Technical Area

Next to the field, along one of its longer sides, there is the technical area. It must contain a computer with two monitors for the purpose of sending GameController messages to the robots and observing compliance with the wireless network usage rules. In addition, there can be a larger monitor facing the field showing the Game State Visualizer.

The technical area can also contain cameras directed at the field and related equipment with the following use cases:

- Stream video data to the Internet to allow the public to watch games
- Record video data for later analysis and creation of statistics

1.7 技术区域

在场地旁边，沿着其中一条较长的边，是技术区域。它必须包含一台带有两个显示器的计算机，用于向机器人发送GameController消息并观察无线网络使用规则的遵守情况。此外，还可以有一个面向场地的较大显示器，用于显示Game State Visualizer。

技术区域还可以包含指向场地的摄像机和相关设备，用于以下场景：

- 将视频数据流到互联网，让公众观看比赛
- 录制视频数据以供后续分析和统计

2 Robot Players

A match is played by two teams, each with a **maximum** of 5 or 7 players, depending on the competition rules. At most one player per team on the field may be designated as *goalkeeper*, the others are all *field players*. When playing at full strength, a team must have a *goalkeeper* on the field.

In addition, each team may prepare *substitute players* outside of the field. A *substitute player* may be substituted in to become a *field player* or *goalkeeper*.

Each of the players has a unique jersey number from the set $\{1, 2, 3, \dots, 20\}$.

2.1 Hardware

All teams must use black, gray, red, blue, or orange plated NAO humanoid robots manufactured by Aldebaran.

Absolutely no modifications or additions to the robot hardware are allowed. No additional hardware is permitted including off-board sensing or processing systems. Additional sensors besides those originally installed on the robots are likewise not allowed. The only exceptions are:

- Setting the passive wrist joints to a fixed position either with glue or a transparent or white duct tape.
- Protecting the fingers with white finger protectors provided by the manufacturer or with transparent or white duct tape.
- Placing white duct tape over the battery case and screw (under the robot jersey) to keep the battery case in place and prevent the battery becoming disconnected.
- A memory stick may remain in the head during operation. Only ordinary USB flash memory keys that sit flush or recessed to the head casing may be utilized. Other USB dongles or devices, as well as memory sticks that are not flush or recessed, are not permitted.

2.2 Team Markers

Robots use colored jersey shirts as team markers. Each jersey shirt has a player number (1–20) printed on it. The team markers are worn as shown in Figure 5.

Teams may use any jersey that was approved for a RoboCup SPL competition in the last 4 years without needing to approve it in 2025 again.

Teams may design and manufacture their own jerseys in any color (multi and many color jerseys are acceptable), but must follow these guidelines:

2 机器人球员

一场比赛由两支队伍进行，每队最多有5名或7名球员，具体取决于比赛规则。每队最多只能有一名球员被指定为守门员，其余都是场内球员。当球队以全阵容比赛时，场内必须有一名守门员。

此外，每支队伍可以在场外准备替补球员。替补球员可以替换上场成为场内球员或守门员。

每位球员都有一个来自 $\{1, 2, 3, \dots, 20\}$ 集合的唯一球衣号码。

2.1 硬件

所有队伍必须使用由 Aldebaran 制造的黑色、灰色、红色、蓝色或橙色涂层的 NAO 人形机器人。

绝对不允许对机器人硬件进行任何修改或添加。不允许额外的硬件，包括场外感知或处理系统。同样不允许在机器人上安装原本未安装的额外传感器。唯一的例外是：

- 将被动手腕关节固定在某个位置，可以使用胶水或透明或白色的胶带。
- 用制造商提供的白色指套保护手指，或使用透明或白色的胶带。
- 在电池盒和螺丝（机器人球衣下方）上贴上白色胶带，以固定电池盒并防止电池断开连接。电池盒。
- 内存棒可能在操作时留在头部。只有与头部外壳平齐或嵌入其中的普通USB闪存盘才能使用。其他USB转接器或设备，以及不平齐或未嵌入的内存棒均不允许使用。

2.2 队标

机器人使用彩色连帽衫作为队标。每件连帽衫上都印有球员号码（1-20）。队标按照图5所示佩戴。

队伍可以使用在过去4年内获得RoboCup SPL比赛批准的任何连帽衫，而无需在2025年再次批准。

队伍可以自行设计和制造任何颜色的球衣（多色和多种颜色的球衣均可接受），但必须遵循以下指南：

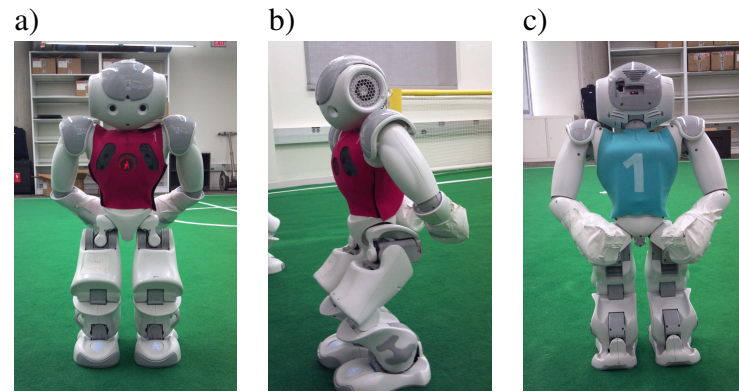


Figure 5: Team markers. a) Front view. b) Side view. c) Back view.

- Jerseys should be the **tank top** style used at RoboCup 2013/2014 and should cover approximately the same areas of the robot as shown in Figure 5. The torso LED must be clearly visible. Jerseys may include the sonar panel used in the 2013/2014 jerseys, although this is not required. Jerseys may not cover the shoulders of the robots.
- Jerseys must have a primary color that comprises more than half of the jersey. The primary color must be recognizable from the front and back.
- Jerseys should not contain distractors, such as large pictures of SPL balls or white stripes on green jerseys.
- A team must wear the jerseys that it starts a game in for the entire game.
- Jersey material must be non-reflecting and non-shiny. Material that is glittery is also not appropriate. Jerseys may be manufactured from fabric and fine mesh.
- Jerseys should be numbered 1–20 on both sides. The numbers must be large and **easily** recognized by humans.
- Teams must have two sets of jerseys for *field players* that are significantly different in terms of their primary color.
- Teams must have two sets of jerseys for *goalkeepers* that are significantly different in terms of their primary color. One of the *goalkeeper* primary colors must be different from both primary colors used for *field players*.
- Designs must be submitted to rc-spl-tc@lists.robocup.org for approval by 2025-05-05. If the team has jersey prototypes, they should submit close-up images of a robot wearing the jersey—these images should be taken from front, back, and side angles. If the team has no prototypes, then designs depicting the expected jersey should be submitted. If submissions show separated front and back halves of jerseys then the team must specify which halves are matched to form home and away jerseys. All images and designs should be submitted in pdf or jpg format.

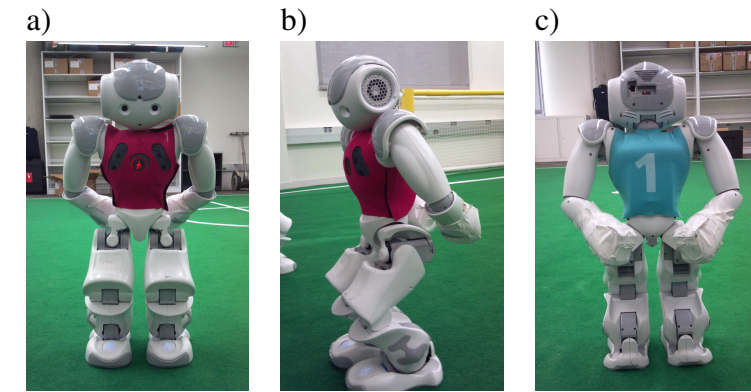


图5：队伍标记。a) 正视图。b) 侧视图。c) 背视图。

- 球衣应为2013/2014年RoboCup使用的背心款式，且应覆盖机器人约相同区域，如图5所示。躯干LED必须清晰可见。球衣可包含2013/2014年球衣使用的声纳面板，尽管这不是必需的。球衣不得覆盖机器人的肩膀。
- 球衣必须有一种主色，其面积超过球衣总面积的一半。主色必须从前方和后方均可识别。
- 球衣不应包含干扰物，例如SPL球的大图片或绿色球衣上的白色条纹。
- 一个队伍必须在整场比赛中穿着它在比赛中开始时穿的球衣。
- 球衣材料必须是非反射且无光泽的。连帽衫可以由织物和细网眼制成。
- 球衣应在两面都编号1-20。号码必须足够大且易于人类识别。被人类识别。
- 球队必须为场内球员准备两套主色调差异显著的球衣。主色调差异显著的球衣。
- 球队必须为守门员准备两套主色调差异显著的球衣。其中守门员的主色调必须与场内球员使用的两种主色调都不同。
- 设计图须于2025-05-05前提交至rc-spl-tc@lists.robocup.org以供批准。若球队有球衣原型，应提交机器人穿着球衣的特写照片——这些照片需从正面、背面和侧面角度拍摄。若球队没有原型，则应提交描绘预期球衣的设计图。若提交的设计图显示球衣前后两半分离，则球队必须明确说明哪两半配对形成主场和客场球衣。所有图片和设计图须以pdf或jpg格式提交。



Figure 6: Example Sponsor/Team Logo placement on legs.



图6：示例赞助商/团队标志在腿部的放置位置。

The jersey sets that are used in a game are decided by the head referee in accordance with the teams during the pre-game meeting (see Section 5.4), subject to the following conditions:

- Both teams' *field player* colors must be easily distinguishable from each other.
- Both teams' *goalkeeper* colors must be easily distinguishable from both teams' *field player* colors.
- The two *goalkeeper* colors may only be similar when this is the case for all choices among the available jersey sets.

The head referee should respect the teams' wishes unless they violate the above conditions, in which case the head referee must bring about a solution.

Some teams wish to include additional information or logos on their robots. The following are allowable:

- Adding sponsor or team logos to the upper legs of the robots (see Figure 6). A box drawn around the non-white area of any logos must not cover more than a 25 cm^2 area.
- Adding small black and white stickers to the torso and/or head of the robots stating the name of the robot, the name of the team, or similar information. These stickers must be small and mostly white.

2.3 Goalkeeper

The *goalkeeper* may use any of the allowed jersey numbers. The *goalkeeper* must wear a jersey with a primary color different from the primary colors used by the *field players* of both teams.

比赛中使用的连帽衫套装由主裁判在赛前会议上根据各队决定（参见第5.4节），但需符合以下条件：

- 两队的场内球员颜色必须容易区分。
- 两队的守门员颜色必须容易区分于两队的场内球员颜色。
- 只有在所有可选球衣套装的选择中都适用时，两个守门员颜色才可能相似。

{v1}主裁判应尊重各队的意愿，除非他们违反了上述条件，在这种情况下，主裁判必须寻求解决方案。

有些队伍希望在他们的机器人上添加额外信息或标志。以下是可以接受的：

- 将赞助商或团队标志添加到机器人的上腿（见图6）。必须绘制一个框围绕任何标志的非白色区域，该框不能覆盖超过 25 cm^2 的区域。
- 在机器人的躯干和/或头部添加小黑白贴纸，标明机器人的名称、团队名称或类似信息。这些贴纸必须很小，且大部分为白色。

2.3 守门员

守门员可以使用任何允许的球衣号码。守门员必须穿着一件主色调与两队场上球员使用的主色调不同的球衣。

2.4 Communications

The robots must play without human control. Communication is only allowed among robots on the field and between the robots and the GameController.

2.4.1 Non-wireless Communications

In general there are no restrictions on communication between robots in play on the field using visual signalling (e. g. gestures) or the robot's built-in microphones, speakers, and infrared transceivers. However, communication that causes excessive discomfort to an audience, affects the safety of an audience, or violates normal playing rules is not permitted.

2.4.2 Wireless Communications

The only wireless hardware allowed to be used by the teams are the wireless network cards built into the NAOs, and the access points provided by the event organizers. All other wireless hardware must be deactivated. A team may be disqualified if one of the team members violates this rule.

Each team will get a range of IP addresses that can be used both for their robots and their computers. The network configuration (e. g. IP addresses, channels, SSIDs, and required encryption) of the fields will be announced at the competition site.

Wireless robot-to-robot communication among the robot players is allowed, as long as it uses the access points provided by the event organizers (using the so-called ad-hoc mode is prohibited), messages are sent via UDP broadcast, and each message's payload size does not exceed 128 B. Each team will be allocated a single UDP port on which to send/receive their team messages. Specifically, a team's port will be 10000 plus that team's GameController number. Unicast communication between robots is prohibited.

The amount of data transmitted by a team in a single game is limited. The limit is measured as the total number of UDP packets sent by any robots of a team. A team may not exceed 1200 packets per game. For each minute of irregular extra time (see Section 3.15), the limit is increased by 60 packets. However, this does not apply to a team that has already exceeded its limit before. The GameController tracks the number of messages that have already been sent and includes counters for packets remaining per team in `RoboCupGameControlData`.

If a team exceeds their limit the game is scored with 0 goals for the offending team. Robot-to-robot communication that violates the SPL rules result in a game scored with 0 goals for the offending team (even when discovered after the game was finished). Messages are counted towards the limit during the `standby`, `ready`, `set` and `playing` states. Limits do not apply during penalty kick shoot-out. If a knock-out game results in a draw and both teams exceeded their message limit, the team last exceeding their limit wins the game.

2.4 通信

机器人必须无人控制地比赛。通信仅允许在场上的机器人之间以及机器人和GameController之间进行。

2.4.1 非无线通信

通常情况下，在场上的机器人之间使用视觉信号（例如手势）或机器人的内置麦克风、扬声器和红外收发器进行通信不受限制。然而，任何引起观众过度不适、影响观众安全或违反正常比赛规则的通信均不被允许。

2.4.2 无线通信

允许团队使用的唯一无线硬件是集成在NAO机器人中的无线网卡，以及活动组织者提供的接入点。所有其他无线硬件必须被禁用。如果团队成员违反此规则，该团队可能会被取消资格。

每个团队将获得一系列可用于其机器人和计算机的IP地址。比赛场地的网络配置（例如IP地址、信道、SSID和所需加密）将在比赛现场公布。

机器人选手之间允许无线通信，只要使用活动组织者提供的接入点（禁止使用所谓的ad-hoc模式），消息通过UDP广播发送，且每条消息的有效载荷大小不超过128字节。每个队伍将被分配一个UDP端口用于发送/接收其队伍消息。具体来说，一个队伍的端口将是10000加上该队伍的GameController编号。机器人之间的单播通信是被禁止的。

一个队伍在一局游戏中传输的数据量有限制。该限制以队伍所有机器人发送的UDP数据包总数来衡量。一个队伍每局游戏不能超过1200个数据包。对于每分钟的非正常额外时间（参见第3.15节），限制会增加60个数据包。但是，这不适用于已经超过其限制的队伍。GameController跟踪已经发送的消息数量，并在RoboCupGameControlData中包含每个队伍剩余数据包的计数器。

如果一个队伍超过其限制，比赛将以0分判给违规队伍。违反SPL规则的机器人之间通信会导致比赛以0分判给违规队伍（即使是在比赛结束后才发现）。在待机、准备、设置和比赛状态下，消息会计入限制。在点球大战期间，限制不适用。如果淘汰赛结果是平局且两队都超过了消息限制，那么最后超过限制的队伍将赢得比赛。

The limit of allowed packets will be lowered in future competitions to encourage smart event-based communication.

In addition to robot-to-robot communication, robots may send:

- Additional status update packets that are sent to the GameController.
- Team specific debug information may be sent to an external computer owned by the team. A robot may send debug information at most once every 2 s in a single UDP packet.

These additional packets do not count towards the team's data limit and may not be used for robot-to-robot communication. They must be sent as unicast and may not be sent as broadcast.

Teams and their robots must not listen into another team's communication.

Robots are not allowed to be connected to access points of fields that are currently running official games of other teams. Robots may only communicate on fields that are not running an official game or fields which they are playing on.

The GameController will use UDP to connect to the robots. The source distribution of the GameController provides the header file *RoboCupGameControlData.h* that defines all messages sent by the GameController to the robots. They correspond to the *robot states* described in Section 3.2.

Robots send status updates (defined in *RoboCupGameControlData.h*) to the GameController. These return packets must be addressed directly to the GameController PC (i.e. not broadcast) and sent on the GameController return UDP port specified by the symbol `GAMECONTROLLER_RETURN_PORT` in *RoboCupGameControlData.h*. The rate at which return packets may be sent is limited to 2 s^{-1} .

The use of remote processing/sensing is prohibited.

未来比赛中，允许的包数量限制将被降低，以鼓励基于事件的智能通信。

除了机器人之间的通信外，机器人还可以发送：

- 向游戏控制器发送额外的状态更新数据包。
- 团队特定的调试信息可能会发送到团队拥有的外部计算机。单个UDP数据包中，一个机器人最多每2秒发送一次调试信息。

这些额外的数据包不计入团队的数据限制，并且不能用于机器人之间的通信。它们必须作为单播发送，不能作为广播发送。

团队及其机器人不得监听其他团队的通信。

机器人不允许连接正在运行其他团队官方比赛的比赛区域接入点。机器人只能在未运行官方比赛的比赛区域或它们正在参与的比赛区域进行通信。

游戏控制器将通过UDP连接到机器人。游戏控制器的源代码分布提供了头文件 *RoboCupGameControlData.h*，该文件定义了游戏控制器向机器人发送的所有消息。这些消息对应于第3.2节中描述的机器人状态。

机器人将状态更新（在*RoboCupGameControlData.h*中定义）发送给游戏控制器。这些返回数据包必须直接地址游戏控制器PC（即不广播），并通过 *RoboCupGameControlData.h*中由符号`GAMECONTROLLER_RETURN_PORT`指定的游戏控制器返回UDP端口发送。返回数据包的发送速率限制为每2秒⁻¹一次。

禁止使用远程处理/传感。

3 Game Process

3.1 Structure of the Game

A game consists of four parts, pre-game meeting (see Section 5.4), the first half, a half-time break, and the second half. Each half is 10 min counted from the initial kick-off. The half-time break is 10 min, and during this time both teams may change robots, change programs, or do anything else that can be done within the time allotted.

The head referee signals the commencement of each half with a single whistle blow (that is, the Initial kick-off, see Section 3.6). The head referee signals the end of the first half with two short whistle blows, and the end of the second half with two short plus one long whistle blow. The head referee should make *all* of these whistle sounds from the T-junction of the halfway line.

3.2 Robot States

Robots can be in *nine* different *primary* states (see Figure 7). Wireless connection must be available, so these states will be set by the GameController. Teams must implement code to receive and correctly respond to wireless GameController packets, and also give a visual indication of the game state.

The usage of the button interface as a replacement for any GameController commands/transitions is not allowed in the main competition!

Should, on both teams, at least two robots have problems with the wireless network or GameController connection the head referee should issue a referee timeout (see Section 3.14). If fewer robots do not respond to the GameController then they are, at the beginning, not included in the game (via a ‘Request for Pick-up’, see Section 3.12), and the game starts without the offending robot. The same applies to robots that no longer respond to the GameController at a later point in the game. Not responding means in this case that they are failing to react to the next GameController signal, such as a penalty or a change in game state.

The primary states are:

Unstiff. It helps to facilitate a consistent and safe handling of the robots for remote competition. During any state, if all head buttons are pressed at least one second, the robot should move to a safe seated/crouched position and unstiffen all joints. So while in the `unstiff` state the robot is not allowed to move in any fashion! After booting, the robots are in their `unstiff` state. Pressing the chest button once while in the `unstiff` state, permits the robot to stiffen its joints and return to the `initial` state, or a state as indicated by GameController.

3 游戏流程

3.1 游戏结构

一场比赛由四个部分组成：赛前会议（参见第5.4节）、上半场、中场休息和下半场。每半场从初始开球时算起，时长10分钟。中场休息时长10分钟，在此期间，双方队伍可以更换机器人、更换程序，或做任何在规定时间内可以完成的事情。

主裁判通过一声哨声（即初始开球，见第3.6节）来示意每半场开始。主裁判以两声短哨声示意第一半场结束，以两声短加一声长哨声示意第二半场结束。主裁判应从中线T字路口发出所有哨声。

3.2 机器人状态

机器人可以有九种不同的主要状态（见图7）。必须保证无线连接可用，因此这些状态将由游戏控制器设置。各队必须实现代码来接收并正确响应无线游戏控制器数据包，同时也要以视觉方式显示游戏状态。

在主比赛中，禁止使用按钮界面来替代任何游戏控制器命令/状态转换！

若双方队伍中至少有两台机器人出现无线网络或GameController连接问题，主裁判应发布裁判暂停（参见第3.14节）。如果响应GameController的机器人数量不足，则它们在游戏开始时不会被选入游戏（通过“请求拾拾”方式，参见第3.12节），游戏将不包含该违规机器人。游戏进行中若机器人不再响应GameController，同样适用此规则。这里的“不响应”是指它们未能对下一个GameController信号做出反应，例如罚球或游戏状态变更。

主要状态包括：

解除僵直。这有助于在远程比赛中实现机器人的一致性和安全操控。在任何状态下，如果所有头部按钮至少被按下1秒钟，机器人应移动到安全的坐姿/蹲姿，并解除所有关节的僵直。所以在解除僵直状态下，机器人不允许以任何方式移动！启动后，机器人处于解除僵直状态。在解除僵直状态下按一次胸部按钮，允许机器人使关节变硬并返回初始状态，或返回GameController指示的状态。

Initial. The robots are free to move at teams convenience and humans are allowed to interact with the robots. This state is not limited in time and teams have access to the field. The GameController will activate this state before `standby` (i.e at the beginning of a half and during a timeout). Shortly pressing the chest button will switch the robot to the `penalized` state.

Standby. The robots are not allowed to move in any fashion besides initially standing up and moving their heads. In `standby`, the robots are awaiting a visual indication from the referee which signals the transition to `ready`.

Ready. In this state, the robots walk to their legal positions for kick-off (see Section 3.7) or a penalty kick (see Section 3.10). They remain in this state, until the head referee decides that there is no significant progress, up to a maximum of 45 s for a kick-off and 30 s for a penalty kick. The GameController can activate sub-states for kick-off and penalty kicks.

Set. In this state, the robots stop and wait for kick-off (see Section 3.7) or a penalty kick (see Section 3.10). Illegally positioned robots are `penalized` and placed on the side of the field. Robots are allowed to move their heads and arms or get up if fallen before the game (re)starts, but they are not otherwise allowed to move their legs or locomote in any fashion. If a robot cannot get up, fallen robot is called (see Section 4.6). The penalty time counter is frozen during this state. Note that all penalized robots are left in place (on the side of the field, or in-place for motion in set) and must wait to get unpenalized. The GameController can activate sub-states for kick-off and penalty kicks.

Playing. In the `playing` state, the robots are playing soccer. Shortly pressing the chest button will switch the robot to the `penalized` state. During the `playing` state, the GameController can activate the sub-states for free kicks (see Section 3.9).

Penalized. A robot is in this state when it has been `penalized`. The only allowed movement is standing up after falling. Any other movement (including turning the head) is disallowed. Shortly pressing the chest button will switch the robot back to the `playing` state.

Finished. This state is reached when a half is finished.

Calibration. This state denotes the robot is acting with automatic calibration. This state may only be entered from `initial` by first pressing the front head button concluded by the chest button, for at least one second.

At the scheduled time, the referee will announce that the transition from the `initial` to `standby` will be executed 60 seconds later. The referee shall call “Standby in 60 seconds” The GameController operator will announce the transition from `initial` to `standby` when the head referee is calling “Standby”. If all humans from playing teams have left the field, the head referee can announce the transition earlier.

Once the game controller operator has confirmed the `standby` state, the referee should wait a random time between 10 to 120 seconds before announcing the transition to `ready` (see Section 5.6.1).

初始状态。机器人可以自由移动，方便队伍操作，人类也可以与机器人互动。此状态无时间限制，队伍可以进入场地。GameController将在待机状态之前激活此状态（即半场开始和暂停期间）。短按胸部按钮将使机器人切换到受罚状态。

待机状态。机器人不允许以任何形式移动，只能最初站立并移动头部。在待机状态下，机器人等待裁判发出视觉信号，以指示过渡到准备状态。

准备状态。在此状态下，机器人走向开球区域（见第3.7节）或罚球区域（见第3.10节）。它们将保持此状态，直到主裁判认为没有显著进展，最长不超过45秒（开球）或30秒（罚球）。游戏控制器可以为开球和罚球激活子状态。

就位状态。在此状态下，机器人停止并等待开球（见第3.7节）或罚球（见第3.10节）。位置非法的机器人将被处罚并放置在场边。在比赛（重新）开始前，机器人允许移动头部和手臂，或如果倒下则站起来，但除此之外不允许移动腿部或以任何形式移动。如果机器人无法站起来，倒下的机器人被称为（见第4.6节）。在此状态下，处罚时间计数器冻结。请注意，所有受处罚的机器人都停留在原地（在场边，或在就位状态下保持位置），必须等待解除处罚。游戏控制器可以为开球和罚球激活子状态。

正在播放。在播放状态下，机器人正在踢足球。短按胸部按钮会将机器人切换到受罚状态。在播放状态下，GameController可以激活任意门的子状态（见第3.9节）。

受罚。当机器人受罚时，它会处于此状态。唯一允许的动作是摔倒后站起来。任何其他动作（包括转动头部）都是不允许的。短按胸部按钮会将机器人切换回播放状态。

结束。当半场结束时，会进入此状态。

校准。此状态表示机器人正在进行自动校准。此状态只能从初始状态进入，方法是先按前头部按钮，然后按胸部按钮，持续至少一秒钟。

在预定时间，裁判将宣布从初始状态到待机状态的转换将在 60 秒后执行。裁判应喊出“60 秒后待机”。GameController 操作员将在主裁判喊出“待机”时宣布从初始状态到待机状态的转换。如果所有参与播放的队伍人员都已离开场地，主裁判可以提前宣布转换。

一旦游戏控制器操作员确认了待机状态，裁判应在宣布过渡到ready状态前等待10至120秒之间的随机时间（参见第5.6.1节）。

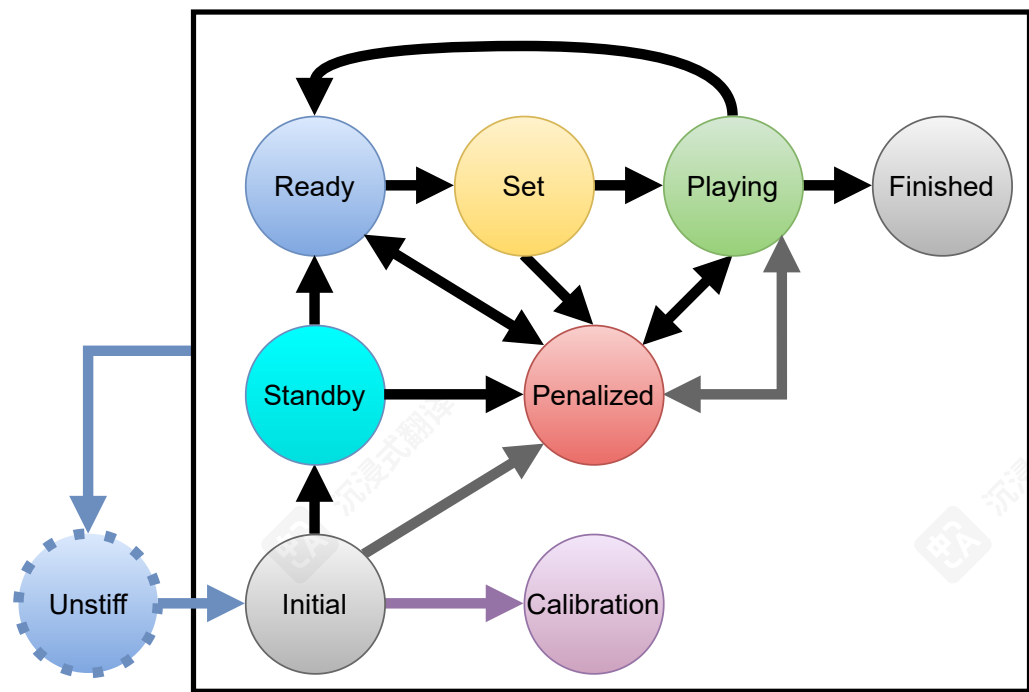


Figure 7: Diagram of the robot states.
Chest button transitions are shown as gray arrows. However, any transition possible should be sent by the GameController.
GameController transitions are shown as black arrows.
 Calibration transitions are shown as purple arrows which mean pressing the **front head button + the chest button**.
 From any state it can be transitioned to the `unstiff` state, shown as a blue arrow, via pressing all **three head buttons**. Pressing the **chest button** in the `unstiff` state allows a transition to the `initial` state, or a state as indicated by the GameController.

Teams should expect that the delay before the referee signals the transition from `standby` to `ready` will be different every time. To announce the transition from `standby` to `ready` state, the referee will raise both hands over their head (see Figure 8). The referee must stand on the opposite side of the technical area behind the touchline and in line with the halfway line. The signal must be held for 2s. The GameController signal for the transition from `standby` to `ready` will be delayed by 45s for Champions Cup and by 35s for Challenge Shield.

The referee will announce the start of the `playing` state with a single whistle blow. The GameController `playing` signal will be delayed by 15s. This delay applies to both kick-off and penalty kicks.

The current game state should be displayed on the LED in the torso. The colors corresponding to the game states are:

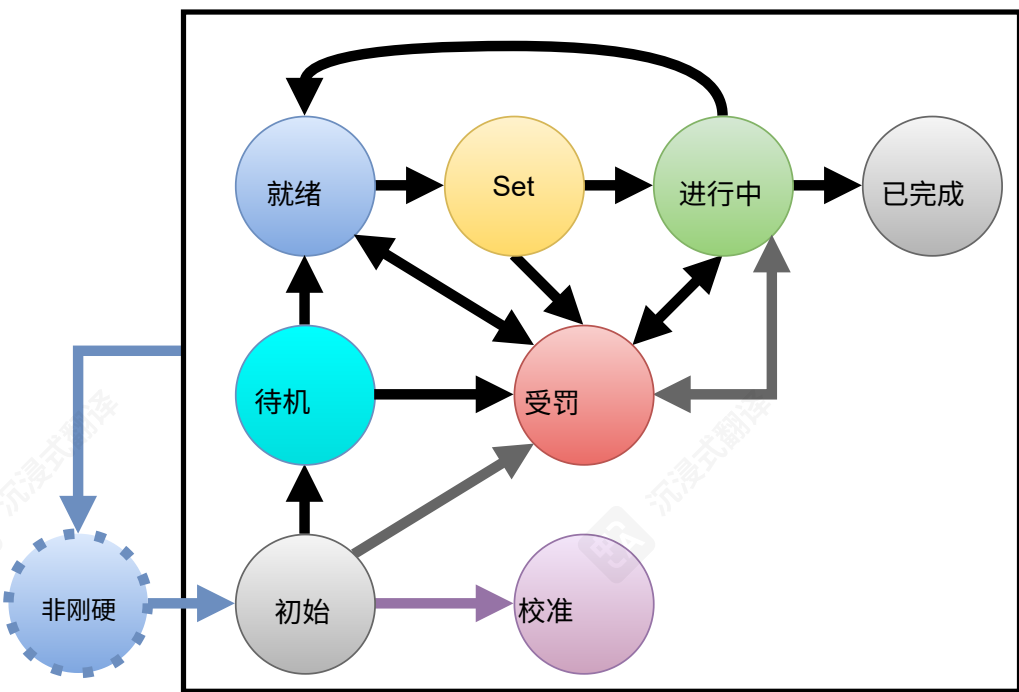


图7：机器人状态图。
 胸按钮的转换用灰色箭头表示。但是，任何可能的转换都应该发送由GameController执行。
 游戏控制器转换用黑色箭头表示。
 校准转换用紫色箭头表示，这意味着按下前头按钮 + 胸口的按钮。
 在任何状态下，都可以通过按下所有三个头部按钮，以蓝色箭头所示的方式过渡到非僵硬状态。
 在非僵硬状态下按下胸口的按钮，可以过渡到初始状态，或由GameController指示的状态。

各队应预期裁判从准备状态发出过渡到就绪状态的信号时会有不同的延迟。裁判将通过将双手举过头顶来宣布从准备状态过渡到就绪状态（见图8）。裁判必须站在技术区域后方、边线对面且与中线对齐的位置。该信号必须保持2秒。GameController从准备状态过渡到就绪状态的信号将延迟45秒（欧冠）和35秒（挑战盾）。

裁判将通过一声哨声宣布比赛状态的开始。GameController比赛信号将延迟15秒。此延迟适用于开球和点球。

当前游戏状态应显示在躯干的LED上。游戏状态对应的颜色是：



Figure 8: Example of referee executing ready signal.

- Unstiff: Blue-Blinking
- Initial: Off
- Standby: Cyan
- Ready: Blue
- Set: Yellow
- Playing: Green
- Penalized: Red
- Finished: Off
- Calibration: Purple

The current GameController requires robots to know both their team number and their robot number within the team. It is each team's responsibility to make sure this is correctly configured. It is recommended that the robot indicates its number within the team on boot up so that this can be easily checked at the start of the game.

3.3 Goal

A goal, including own goal, is achieved when the entire ball (not only the center of the ball) goes over the goal-side edge of the goal line, i. e. the ball is completely inside the goal.¹

¹The goal line is part of the field.



图8：裁判执行准备信号的示例。

- 未僵硬：蓝色闪烁
- 初始：关闭
- 待命：青色
- 准备：蓝色
- 就位：黄色
- 比赛：绿色
- 惩罚：红色
- 完成：关闭
- 校准：紫色

当前的GameController要求机器人既要知道自己的队伍编号，也要知道队伍内的机器人编号。确保正确配置是每个队伍的责任。建议机器人在启动时显示其队伍内的编号，以便在比赛开始时轻松检查。

3.3 目标

一个进球，包括自摆球，当整个球（而不仅限于球心）越过球门线的球门侧边缘时即被判定进球，也就是说球完全在球门内。¹

¹目标线是场地的一部分。

The head referee signals a goal by a single whistle blow, followed by the call “Goal <color>”. The head referee should point with one arm towards the center of the field. To assist robots listening for whistles, the referee should blow the whistle from on the carpet at the end of the fields where the goal was scored.

After a team scores a goal, the game proceeds with a kick-off (see Section 3.7) for their opponents. The GameController signal (to the robots) of a goal being scored, will be delayed by 15 s.

3.3.1 Invalid Goal

A goal is invalid (that is, it can never be awarded) in the following circumstances:

1. When an indirect kick is required but has not occurred (see Section 3.4).
2. When the last contact of the ball was with an attacking robot that played the ball with the arms/hands as defined in Section 4.11. However, an own goal may be scored by any defending robot playing with arms/hands.
3. When a team scores on themselves and there are no opponent robots on the field that are active (a definition of *active* is given in Section 4.6).

In these cases a goal is not scored (that is, the goal is ruled invalid), and the game will proceed with a Goal Kick (see Section 3.9). The head referee should also advise why the goal is invalid, such as by calling “Not indirect”, “Played with hands” or “No own goal”.

3.4 Indirect Kick

In Champions Cup and Challenge Shield, both teams may only score a goal through two distinct ball contacts involving different robots within its own team following any affected start or restart in the playing phase, regardless of the initiator of the set play/free kick:

- A robot may not score a goal from a direct kick, including via deflections.
- Set plays considered to be indirect are Kick-off, Kick-in, Pushing free kick, Corner kick and Goal kick or after an unsuccessful Penalty kick.
- The ball must be deliberately played-at a second time by another robot of its own team before a goal may be scored. A deliberate play at the ball includes successfully kicking the ball, dribbling the ball or the goalkeeper playing at the ball with its hands.
- If an indirect free kick is kicked directly into the opponent’s goal, the goal does not count and a Goal kick is awarded.

主裁判通过一声哨声示意进球，随后喊出“进球 <颜色>”。主裁判应单臂指向场地中央。为协助机器人监听哨声，裁判应在进球的场地末端地毯上吹哨。

球队进球后，比赛将进行一次开球（参见第3.7节），为对方球队进行。GameController向机器人发送进球信号时，会有15秒的延迟。

3.3.1 无效进球

在以下情况下，进球无效（也就是说，它永远不可能被判罚）：

1. 当需要间接开球但尚未发生时（参见第3.4节）。
2. 当球的最后一次接触是攻方机器人，且该机器人按照第4.11节的规定用手臂/手触球时。然而，任何防守机器人用手臂/手触球，也可能进球得分。
3. 当一个球队自己进球，且场上没有对方机器人时活跃（活跃的定义在4.6节中给出）。

在这种情况下，不会得分（也就是说，进球被判定无效），比赛将继续进行角球（见3.9节）。主裁判还应该说明为什么进球无效，例如通过喊“不是间接球”、“用手触球”或“没有乌龙球”。

3.4 间接球

在冠军杯和挑战盾赛中，在比赛阶段中，任何受影响的开始或重新开始之后，双方队伍都只能通过涉及不同机器人且发生在己方队伍内的两次不同球触来得分，无论发球方是谁：

- 机器人可能无法通过直接射门得分，包括通过折射的方式。
- 被认为是间接球的比赛包括开球、界外球、推击自由球、角球和门球或在一次未成功的点球之后。
- 必须由其队友故意再次触球，才能进球。故意触球包括成功踢球、带球或门将用手触球。
- 如果间接任意球直接踢入对方球门，进球无效，将判给球门球。

- Note that an own-goal may always be scored without requiring an indirect kick.

There are two exceptions to this rule:

- A direct kick is allowed for the duration of the penalty kick, but this rule still applies after this kick.
- If a team is awarded a free kick but fails to execute it within the allotted time, the opposing team may attempt to score directly if they play the ball before the team who was originally awarded the free kick does. “Execute” is intended as “kick or move the ball clearly”, as per section Section 3.9.2.

In both cases, the direct goal is only counted if the kicking robot played the ball once. The indirect kick rule returns into effect when the ball comes to a full stop after the kick, or if the same robot plays the ball again before then. In this case, any contacts made by the kicking robot, or any other robot, during the direct kick count towards the indirect kick rule.

Example 1: Player 2 (of the red team) kicks the ball to Player 3 (of the red team), who then kicks the ball into the blue team’s goal. This is a successful indirect kick, and the goal counts.

Example 2: Player 2 (of the red team) kicks the ball at the goal, and it is deflected off the side of the foot of a blue team robot into the goal. This is *not* an indirect kick, the goal does not count and a Goal kick is awarded to the blue team.

Example 3: Player 2 (of the red team) kicks the ball “upfield”. A blue team robot kicks the ball a short distance, after which Player 2 kicks the ball again into the blue team’s goal. This is *not* an indirect kick, the goal does not count and a Goal kick is awarded to the blue team.

Example 4: Player 2 (of the red team), walks up and dribbles the ball. Player 2 then stops, and visibly backs away from the ball, before approaching to dribble a *second* time. The robot then scores. This is *not* an indirect kick, the goal does not count and a Goal kick is awarded to the blue team.

Example 5: Player 2 (of the red team) kicks the ball “upfield”. A blue-team robot kicks the ball again into the red team’s goal. This is *not* an indirect kick, the goal does not count and a Goal kick is awarded to the red team.

Example 6: Player 2 (of the red team) is awarded a Penalty kick and kicks the ball towards the blue team’s goal. The goalkeeper blocks the ball. The referee then calls “Ball Free”. Player 2 (of the red team) kicks the ball again and scores. This is *not* an indirect kick, the goal does not count and a Goal kick is awarded to the blue team.

Example 7: The blue team is awarded a free kick, but doesn’t play the ball. The timer expires, and Player 2 (of the red team) kicks directly into the blue goal. The failed-free-kick exception applies, and the goal counts.

- 请注意，即使不需要间接射门，也可能进球乌龙。

这条规则有两个例外：

- 在罚球过程中允许直接射门，但这个规则在这个踢。
- 如果一个队伍被判罚任意球但未在规定时间内执行，如果对方队伍在原任意球被判罚的队伍之前触球，则对方队伍可以尝试直接得分。“执行”指的是“踢球或明显移动球”，根据第3.9.2节的规定。

在这两种情况下，直接进球只有在踢球的机器人触球一次时才会计数。当球在射门后被完全静止，或同一机器人在此之前再次触球时，间接球规则会重新生效。在这种情况下，踢球机器人或任何其他机器人在直接射门过程中的任何接触都将计入间接球规则。

示例1：红队的2号球员将球踢给红队的3号球员，然后3号球员将球踢入蓝队的球门。这是一个成功的间接球，进球有效。

示例 2：2 号球员（红队队员）将球踢向球门，球被蓝队机器人脚边侧面挡出并进入球门。这不是间接界外球，进球无效，蓝队获得球门球。

示例 3：2 号球员（红队队员）将球“向上场踢”。蓝队机器人将球短距离踢出后，2 号球员再次将球踢入蓝队球门。这不是间接界外球，进球无效，蓝队获得球门球。

示例 4：2 号球员（红队队员）上前走动并控球。2 号球员随后停下，明显远离球，然后再次接近准备控球。机器人随后进球。这不是间接界外球，进球无效，蓝队获得球门球。

示例 5：2 号球员（红队队员）将球踢向“球场上方”。一名蓝队机器人再次将球踢入红队球门。这不是间接界外球，进球无效，红队获得球门球。

示例 6：2 号球员（红队队员）获得点球，将球踢向蓝队球门。守门员挡球。裁判随后喊出“球自由”。2 号球员（红队队员）再次踢球并得分。这不是间接界外球，进球无效，蓝队获得球门球。

示例 7：蓝队获得任意球，但没有触球。计时器到时，红队的 2 号球员直接将球踢入蓝队球门。触发未执行任意球异常，进球有效。

Example 8: The blue team is awarded a free kick, but doesn't play the ball. The timer expires, and Player 2 (of the red team) kicks the ball 1 meter away. The ball stops, Player 2 kicks again, and scores. The exception does *not* apply for the red team because Player 2 played the ball twice. The goal does not count and a Goal kick is awarded to the blue team.

Example 9: The blue team is awarded a free kick, but doesn't play the ball. The timer expires, and Player 2 (of the red team) passes the ball to Player 3 (of the red team), who kicks again and scores. Although the exception doesn't apply because Player 3 played the ball as well, this is still an ordinary indirect kick, so the goal counts.

Example 10: The blue team is awarded a free kick, but doesn't play the ball. The timer expires, and Player 2 (of the red team) kicks the ball 1 meter away. Player 5 (of the blue team) kicks the ball, then Player 2 kicks and scores. The exception does *not* apply for the red team because Player 5 also played the ball. The goal does not count and a Goal kick is awarded to the blue team.

Example 11: The blue team is awarded a free kick, but doesn't play the ball. The timer expires, then Player 5 (of the blue team) kicks the ball. Player 2 (of the red team) then kicks the ball and scores. The exception does *not* apply for the red team because a blue-team robot played the ball first. The goal does not count and a Goal kick is awarded to the blue team.

3.4.1 Fallback Mode for Challenge Shield

This rule aims to allow **Challenge Shield** teams to play games even in the unfortunate case where severe circumstances make them unable to field enough robots to play by the default indirect kick rules, by relaxing the requirements to score a goal if they cannot play with more than two robots.

Each Challenge Shield team participating in a game shall declare their operational capacity for that game by choosing to play either in **normal mode** or in **fallback mode**. If at least one team plays in *fallback mode*, the whole game follows different, more lenient rules for indirect kicks, detailed in Section 3.4.2.

If a team plays in *fallback mode*, their maximum team size is **two** robots. The other robots do not participate in the game, unless substituted as allowed by the rules (see Section 3.12). Note that one team's choice of operational mode does **not** limit the number of robots the opposing team is allowed to play with.

Teams decide which operational mode they play in before the beginning of each half. Furthermore, during any timeout (regardless of who initiated it), each team may choose to switch from *normal mode* to *fallback mode*, but not vice versa. In both cases, teams must inform the head referee and the GameController operator of their choice, and they are encouraged to make this decision as early as possible.

Changing operational modes during normal gameplay is not allowed.

示例 8：蓝队获得任意球，但没有触球。计时器到时，红队的 2 号球员将球踢出 1 米远。球停下后，2 号球员再次射门得分。红队不适用该异常，因为 2 号球员触球两次。进球无效，蓝队获得球门球。

示例 9：蓝队获得任意球，但没有触球。计时器到时，红队的 2 号球员将球传给红队的 3 号球员，后者再次踢球并得分。尽管因为 3 号球员也触球了，这个例外不适用，但这仍然是一个普通的间接任意球，进球有效。

示例 10：蓝队获得任意球，但没有触球。计时器到时，红队的 2 号球员将球踢出 1 米远。蓝队的 5 号球员踢球，然后 2 号球员再次踢球并得分。由于蓝队的 5 号球员也触球了，这个例外不适用于红队。进球无效，蓝队获得球门球。

示例 11：蓝队获得任意球，但没有触球。计时器到时后，蓝队的 5 号球员踢球。红队的 2 号球员随后踢球并得分。由于蓝队机器人首先触球了，这个例外不适用于红队。进球无效，蓝队获得球门球。

3.4.1 挑战盾备用模式

本规则旨在允许挑战盾队伍在不幸遭遇严重情况导致无法部署足够机器人以执行默认间接射门规则时，通过放宽进球要求来参与比赛，前提是他们无法使用超过两台机器人进行比赛。

每支参与比赛的挑战盾队伍都应通过选择正常模式或备用模式来声明其该场比赛的操作能力。如果至少有一支队伍采用备用模式，整场比赛将遵循针对间接射门的更宽松规则，具体细节见第 3.4.2 节。

如果队伍采用备用模式，其最大队伍规模为两台机器人。其他机器人不参与比赛，除非根据规则允许替换（见第 3.12 节）。请注意，一支队伍的操作模式选择不会限制对方队伍允许使用的机器人数量。

队伍应在每半场开始前决定其参与的比赛模式。此外，在任何暂停期间（无论由谁发起），每支队伍都可以选择从正常模式切换到备用模式，但反之则不可。在这两种情况下，队伍必须通知主裁判和游戏控制器操作员其选择，并鼓励尽早做出此决定。

在正常游戏过程中不允许更改操作模式。

The GameController communicates to the robots the operational mode each team plays in, while the TeamCommunicationMonitor and the GameStateVisualizer display this same information for the convenience of the referees.

3.4.2 Indirect Kick in Fallback Mode

As long as at least one team plays in *fallback mode*, **both teams** play by the following rules instead of the default ones for indirect kicks.²

Under these variant rules, after any restart into playing from a free kick, the attacking team may only score a goal via an indirect kick. Set plays considered to be indirect are Kick-in, Pushing free kick, Corner kick and Goal kick or after an unsuccessful Penalty kick.

The conditions for an indirect kick to be valid while at least one team plays in *fallback mode* are as follows:

- A robot may not score a goal from a direct kick, including via deflections.
- The ball must be deliberately played-at a second time (by either another robot, or the same robot) before a goal may be scored. A deliberate play at the ball includes successfully kicking the ball, dribbling the ball (and subsequently leaving possession of the ball), or the goalkeeper playing at the ball with its hands.
- If a robot plays the ball to itself, this means that the ball must leave a circular area of at least 0.75 m around the robot before the ball is played a second time and to be considered as an indirect kick.

Direct kicks are only allowed in the following situations:

- On the first kick at kick-off.
- For the duration of a penalty kick, though the indirect kick rule returns into effect after the set play expires. In this case, any plays made during the penalty kick count towards this rule.
- Note that an own-goal may always be scored without requiring an indirect kick.

Example 1: Player 2 (of the red team) kicks the ball to Player 3, who then kicks the ball into the blue-team’s goal. This is a successful variant indirect kick, and the goal counts.

²For reference, teams who participated in Challenge Shield in 2024 may notice that these are the same indirect kick rules used that year. They have been slightly rephrased to fit their new role, but they are functionally unchanged.

GameController 向机器人传达各队伍所执行的操作模式，而 TeamCommunicationMonitor 和 GameStateVisualizer 则为裁判的方便起见显示相同的信息。

3.4.2 落位模式下的间接射门

只要至少有一支队伍处于落位模式，双方队伍将按照以下规则进行间接射门，而不是间接射门的默认规则。²

在这些变体规则下，从任意一次自由球重新开始比赛后，进攻队伍只能通过间接射门得分。被认定为间接射门的定位球包括：球门球、界外球、角球和门将用手门将球，或一次未成功的点球。

当至少有一支队伍处于落位模式时，间接射门有效的条件如下：

- 机器人不能通过直接射门得分，包括通过折射得分。
- 在得分之前，球必须被故意第二次触球（由另一台机器人或同一台机器人触球）。故意触球包括成功将球踢出、带球（并随后失去球权），或门将用手触球。
- 如果一个机器人将球踢给自己，这意味着球在被第二次踢出前必须离开机器人周围至少0.75米的圆形区域，才能被视为间接犯规。

直接犯规只允许在以下情况下进行：

- 在开球时的第一次触球。
- 在罚球过程中，尽管间接犯规规则在罚球结束后恢复生效。在这种情况下，罚球期间的所有触球都计入该规则。
- 请注意，进球始终可以不被要求间接犯规而直接得分。

示例 1：2 号球员（红队队员）将球踢给 3 号球员，随后 3 号球员将球踢入蓝队的球门。这是一个成功的间接界外球，进球有效。

²参考而言，2024年参加挑战盾赛的球队可能会注意到，这些是当年使用的相同间接犯规规则。它们被稍作修改以适应新角色，但功能上没有变化。

Example 2: Player 2 (of the red team) kicks the ball at the goal, and it is deflected of the side of the foot of a blue-team’s robot into the goal. This is *not* an indirect kick, and the goal does not count.

Example 3: Player 2 (of the red team) kicks the ball “upfield”. A blue-team robot kicks the ball a short distance, after which Player 2 kicks the ball again into the blue team’s goal. This is a successful variant indirect kick, and the goal counts.

Example 4: Player 2 (of the red team), walks up to and dribbles the ball. To be an indirect kick, Player 2 must then stop, and visibly back away from the ball, before approaching to dribble a *second* time. The robot then scores. This is a successful variant indirect kick.

3.5 Field-Side Selection and Initial Kick-off

- The head referee throws a coin with one side for the home team and the other side for the away team according to the schedule. The team that wins the coin toss can either choose which goal to play for in the first half or take the kick-off.
- Depending on the decision above, the opposing team is given either the kick-off or may choose which goal to play for in the first half.
- The team that decided which goal to play for in the first half takes the kick-off at the start of the second half.
- In the half-time break both teams switch the sides of the field at play for the other goal.
- Teams must not change primary jersey colors for field players or goalkeepers during a match.

3.6 Initial Kick-off

The first kick-off at the start of each half or after a timeout is an initial kick-off. Robots have to be ready next to the field at latest 2 min before an initial kick-off. All robots must be in the `unstiff` state.

Latest 1 min before the game starts all robots must be placed on the touchlines facing the other touchline in their own half by their team.

Robots are brought into the `initial` state after they are placed. Once all robots have been placed and all human team members have left the field, the head referee officially starts or resumes the game by calling the beginning of the `standby` state. No further human intervention is allowed by the teams after this point.

Once the robots receive the `ready` visual signal from the referee or the signal from the GameController, whichever happens first, they proceed as described in Section 3.7.

示例 2：2 号球员（红队队员）将球踢向球门，球被蓝队机器人的脚侧面挡入球门。这不是间接界外球，进球无效。

示例 3：2 号球员（红队队员）将球踢向“中场线外”。一个蓝队机器人将球短距离踢出后，2 号球员再次将球踢入蓝队球门。这是一个成功的变通间接射门，进球有效。

示例 4：2 号球员（红队队员）走向球并控球。要成为间接射门，2 号球员必须停下，明显远离球，然后再次接近控球。机器人随后进球。这是一个成功的变通间接射门。

3.5 场边选择和初始开球

- 主裁判根据赛程抛掷硬币，一面代表主队，另一面代表客队。赢得硬币抛掷的队伍可以选择上半场进攻哪个球门，或执行开球。
- 根据上述决定，对方队伍将获得开球权，或可以选择选择上半场为哪个球门踢球。
- 在上半场决定为哪个球门踢球的队伍在下半场开始时获得开球权。下半场。
- 在半场休息时，两队都会交换比赛场地的另一侧，为对方的球门做准备。
- 在比赛中，球队不得更改场内球员或守门员的主要球衣颜色。

3.6 开球

每半场开始时的首次开球或暂停后都是初始开球。机器人在初始开球前至少需在场地旁边就位2分钟。所有机器人必须处于非僵硬状态。

比赛开始前1分钟内，所有机器人必须由其队伍放置在边线上，面向对方半场的边线。

机器人放置后进入初始状态。当所有机器人已放置且所有人类团队成员已离开场地后，主裁判通过呼叫待命状态开始或恢复比赛。此后，各队伍不得再进行任何人为干预。

一旦机器人从裁判或游戏控制器那里接收到准备就绪的视觉信号，以先发生者为准，它们将按照第3.7节所述进行操作。

3.7 Kick-off

For kick-off, the robots run through three states: `ready`, `set`, and `playing`.

In the `ready` state, the robots must walk to legal kick-off positions for their team. Upon reaching the `set` state, the robots must stop and remain stationary until the kick-off is signaled by the referee (see Section 4.5.2). The following restrictions apply from the beginning of the `set` state until the ball comes into play:

- The kick-off team can be positioned within their own half and the center circle. No robot may block the center mark. Up to one robot of the kick-off team is allowed to be outside the own half, provided it is inside the center circle. If more than one robot is outside the own half, only the one closest to the center mark is legal. This robot is then called *kick-off* player and must execute the kick-off. If any other robot of the kick-off team executes the kick-off, the robot wrongly taking the kick-off is considered to be illegally positioned and a moved ball is placed back.
- The defending team can be positioned within their own half excluding the center circle.
- Robots are allowed to be positioned inside their own goal. All other positions that are fully on the border strip are illegal for both teams.
- Both teams are also subject to restrictions on the goal area (see Section 4.4.2).

All robots that do not reach legal positions before the `set` state begins or enter illegal positions before the ball is in play will be penalized with the “Illegal Position” penalty (see Section 4.4.1). An exception applies to robots making reasonable effort to return to the field from a standard penalty and must not be penalized for illegal position.

A referee places the ball on the center mark. If the ball is moved by one of the robots during the `set` state, it is placed back by one of the referees.

The head referee signals the kick-off by a single whistle blow, followed by the call “Playing”. The head referee must signal this from the T-junction of the halfway line. 15 s after the head referee has signaled the kick-off, the robots’ state is switched to `playing` by the GameController (regardless of when the ball comes into play).

3.7.1 Ball in Play

The ball is in play and kick-off ends once

- it is touched by the attacking team and moved clearly or
- 10 s have elapsed in the `playing` state.

The GameController and head referee will indicate this by the call “Ball Free”.

3.7 开球

对于开球，机器人会经历三种状态：准备、准备就绪和比赛。

在准备状态下，机器人必须走到其队伍的合法开球位置。到达设定状态后，机器人必须停止并保持静止，直到裁判发出开球信号（见第4.5.2节）。从设定状态开始直到球开始比赛，适用以下限制：

- 开球队伍可以位于自己的半场和中心圆内。任何机器人不得阻挡中心标记。开球队伍最多允许一台机器人在自己的半场外，前提是它必须在中心圆内。如果一台以上机器人在自己的半场外，只有最靠近中心标记的机器人是合法的。该机器人被称为开球队员，并必须执行开球。如果开球队伍的其他机器人执行了开球，错误开球的机器人将被视为位置非法，并将球放回。
- 防守队伍可以位于自己的半场，但排除中心圆。
- 机器人允许位于自己的球门内。所有完全位于边界条带上的其他位置对双方队伍均属非法。
- 双方队伍在球门区域也受到限制（见第4.4.2节）。

所有在设定状态开始前未到达合法位置或在球进入比赛前进入非法位置的机器人将被处以“非法位置”处罚（见第4.4.1节）。对于从标准处罚中努力返回场地的机器人适用例外情况，不得因非法位置而受到处罚。

裁判将球放置在中心标记处。如果在设定状态期间，球被其中一个机器人移动，则由裁判将其放回。

主裁判通过一声哨声示意开球，随后喊出“比赛开始”。主裁判必须从中线T字路口发出此信号。主裁判示意开球后15秒，游戏控制器将机器人状态切换为比赛状态（无论球何时进入比赛）。

3.7.1 球在比赛进行中

球在比赛进行中，开球结束

- 被进攻方触球并明显移动，或
- 10在比赛状态中经过的时间已满。

The GameController和主裁判将通过“球自由”的呼报来表示这一点。

3.8 Kick-in

A ball is considered to have left the field when there is no part of the ball over the outside of the boundary line (i. e. the line itself is in). If the ball leaves the field it will be replaced on the field by an assistant referee. Balls are deemed to be out based on the team that last touched the ball, irrespective of who actually kicked the ball.

If the ball goes over a touchline then the assistant referee will replace the ball back on the point of that touchline where it went out. A free kick (see Section 3.9) is awarded to the team that did *not* last touch the ball by the referee calling “Kick-in <color>”.

If the ball goes over a goal line then the assistant referee will replace the ball back on the field, depending on which team last touched the ball.

- If the ball was last touched by the defensive team, a *Corner Kick* (see Section 3.9) is awarded to the attacking team. The referee calls “Corner Kick <color>” and the ball is placed on the corner on the same side of the field that the ball was kicked-out.
- If the ball was last touched by the offensive team, a *Goal Kick* (see Section 3.9) is awarded to the defensive team. The referee calls “Goal Kick <color>”, and the ball is placed on the corner of the goal area on the same side of the field that the ball was kicked-out. That is, the corner inside the field, not the T-junction where the goal area meets the goal line.

In these examples, “red half of the field” refers to the half the red team is defending.

Example 1: The red goalkeeper kicks the ball out the end of the field to the right of the goal. The referee calls “Corner Kick blue”, the ball is placed on the corner to the right of the goal and a free kick is started.

Example 2: A blue robot kicks the ball out the end of the field to the right of the goal the red team is defending. The referee calls “Goal Kick red” and the ball is placed on the right corner of the goal area.

Example 3: A blue robot at midfield kicks the ball over the left touchline 2 m into the red half of the field. The referee calls “Kick-in red” and the ball is replaced on the left touchline where it went out.

Example 4: A blue robot kicks the ball but the ball touches a red robot at midfield before leaving the field near the halfway line. The ball is regarded as out by red, the referee calls “Kick-in blue” and the ball is replaced on the touchline where it went out.

3.8 开启

当球完全越过边界线外（即线本身在界内）时，球被视为已出界。如果球出界，将由助理裁判将其重新放回场内。球的出界判定基于最后触球的队伍，而与实际踢球的人无关。

如果球越过边线，则助理裁判会将球放回球出界时的那条边线的相应位置。由最后触球的球队获得任意球（参见第3.9节），裁判通过喊出“开球 <颜色>”来执行。

如果球越过球门线，则助理裁判会将球重新放回场上，具体取决于哪个队最后触球。

- 如果球最后是被防守队触碰的，则攻方将获得角球（参见第3.9节）。裁判员鸣哨示意“角球 <颜色>”，并将球放置在场角，该角与球出界的一侧相同。
- 如果球最后是被攻方触碰的，则判给守方一个球门球（见第3.9节）。裁判员鸣哨并喊“球门球 <颜色>”，同时将球放在球场同侧球门区的角上。也就是说，放在场内角，而不是球门区与球门线相接处的T形接头处。

在这些示例中，“红半场”指的是红队防守的半场。

示例1：红门将在球门右侧将球踢出球场。裁判鸣哨“蓝方角球”，球被放在球门右侧的角旗处，并开始角球。

示例2：一个蓝方机器人将球踢出球场，方向是红队防守的球门右侧。裁判鸣哨“红方界外球”，球被放在球门右侧的角旗处。

示例3：一个蓝方机器人从中场将球踢过左侧边线2米进入红半场。裁判鸣哨“红方界外球”，球被放在球出界的左侧边线处重新投入比赛。

示例 4：一个蓝色机器人踢球，但球在离开场地靠近中线之前碰到了一个红色机器人。红方认为球出界了，裁判喊“蓝方踢球”，球被替换在球出界的边线上。

3.9 Free Kick

A free kick is initiated:

- When the ball goes over the touchlines, termed *Kick-in*.
- When the ball goes over the goal lines initiated by the defensive team, termed *Corner Kick*.
- In place of a goal line Kick-in initiated by the offensive team, also termed a *Goal Kick*.
- When a pushing penalty (see Section 4.10) is awarded near the ball, termed a *Pushing Free Kick*.

The head referee will announce a free kick, calling one of:

1. For a *Kick-in/Goal Kick/Corner Kick*: “Kick-in/Goal Kick/Corner Kick <team>” for the team that did not last touch the ball.
2. For a *Pushing free kick*: “Foul <offending color><offending number>” for the pushing robot.

The GameController will then activate the sub-state for the respective free kick. Note that in the case of the Pushing free kick the sub-state is activated automatically through the “Foul”. The team who is awarded the free kick (termed the attacking team) has 30 s to complete the kick.

3.9.1 Visual gesture

In Champions Cup only, the referee communicates which team is to take the free kick via a visual gesture. The GameController packet will tell the robots which free kick sub-state is active, but not who is to kick the ball.

The gesture consists of raising one arm horizontally and sideways, parallel to the field line, while the other arm remains at rest (see Figure 9). The raised arm is the one pointing towards the defending team’s goal, i.e. the goal of the team that is *not* taking the free kick.³

This gesture is made by the head referee and held for at least 5 seconds after the GameController signal is sent. The head referee is encouraged to seek the acknowledgement of the GameController operator and/or look at the field-facing monitor (if available) in order to determine the timing.

The referee makes an effort to gesture the signal standing as close as possible to the T-junction joining the half-way line to the touchline opposite to the technical area. However, their actual

³For reference, teams who participated in the 2023 Visual Referee Challenge may notice that this the same as the “Kick-in” gesture from back then, but with the opposite arm raised.

3.9 界外球

界外球是这样开始的：

- 当球越过边线时，称为踢球入界。
- 当球越过防守方发起的球门线时，称为角球。
- 当球越过进攻方发起的球门线时，称为门球。
- 当在球附近判罚推人犯规（参见第4.10节）时，称为推人界外球。
界外球。

主裁判将宣布一个任意球，并点名其中一方：

1. 对于界外球/球门球/角球：“界外球/球门球/角球 <队>” 给予最后触球方未触球的队伍
未最后触球的队伍。
2. 对于推人任意球：“犯规 <犯规方颜色><犯规方号码>” 给予推人的机器人
机器人。

游戏控制器随后将激活相应任意球的子状态。注意，在推人任意球的情况下，子状态会通过“犯规”自动激活。获得任意球的一方（称为进攻方）有30秒时间完成射门。

3.9.1 视觉手势

仅在冠军杯中，裁判通过视觉手势传达哪个队伍将踢任意球。游戏控制器数据包会告诉机器人哪个任意球子状态处于激活状态，但不会告知谁将踢球。

该手势包括将一只手臂水平地横向抬起，与场地线平行，而另一只手臂保持静止（见图9）。抬起的手臂指向防守队伍的球门，即不踢任意球的队伍的球门。³

主裁判会做出这个手势，并在游戏控制器信号发送后至少保持5秒钟。鼓励主裁判寻求游戏控制器操作员的确认，并/或查看面向场地的显示器（如果可用），以确定时机。

裁判努力做出手势，信号旗尽可能靠近T字路口，该路口连接中线与远离技术区域的边线。然而，他们的实际

³作为参考，参加2023年视觉裁判挑战的队伍可能会注意到，这与当时的“开球”手势相同，但抬起的手臂相反。

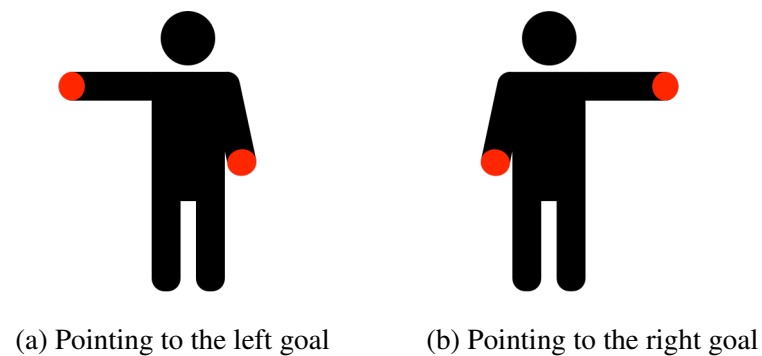


Figure 9: Visual signals for set plays.

positioning along the touchline may deviate slightly from this standard position, based on the current game situation and the timing of the gesture. Teams are not required to scan the entire touchline: in broad terms, a robot standing a few meters away from the T-junction should be able to find the referee in its camera field with no or minimal head motion.

Teams are encouraged to make an effort to familiarize with looking at the referee and interpreting the visual signal even in cases where such an action may not seem strictly necessary, as more information may be communicated in this way rather than via GameController packets in future years.

In Challenge Shield, this visual signal does not need to be performed and the kicking team is communicated by the GameController.

3.9.2 Execution

When necessary, the referee may need to place the ball. For a pushing free kick, the ball will be left in place, and only repositioned in accordance with the pushing rules (see Section 4.10). If the ball left the field, the ball will be positioned as described in Section 3.8.

During the free kick, only the attacking team may approach within 0.75 m to the ball. All robots of the defensive team must immediately move away from the ball. Defensive robots that violate these restrictions are penalized with the “Illegal Positioning” penalty (see Section 4.4) which results in a standard removal penalty (see Section 4.2). Additional penalties against any further robots during the free kick, including Pushing, do not result in an additional free kick, but still use the appropriate removal penalty.

A free kick is deemed completed and play returns to normal if:

- The attacking team moves the ball clearly, except for a robot getting up which is exempt from this rule.

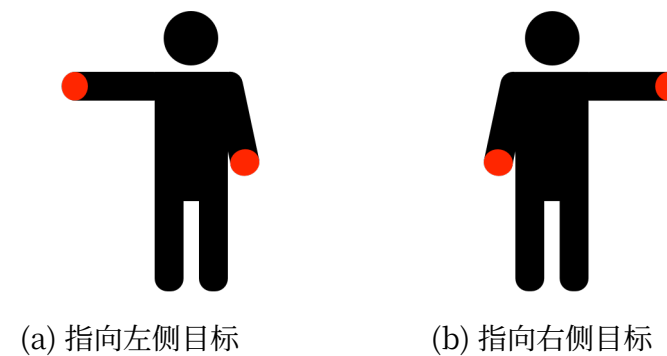


图 9：定位战术的视觉信号。

裁判员努力做出手势，将信号旗尽可能立在T字路口附近，该路口连接中线与远离技术区域的边线。然而，他们实际在边线上的站位可能会略微偏离这一标准位置，具体取决于当前比赛情况和手势的时间。球队无需扫描整条边线：大致而言，若机器人距离T字路口几米远，它应该能在其摄像头视野中找到裁判员，且无需或只需极小的头部动作。

鼓励各队努力熟悉观察裁判并解读视觉信号，即使在某些情况下这种做法似乎并非绝对必要，因为未来可能通过这种方式传递更多信息，而非通过游戏控制器数据包。

在挑战盾赛中，这种视觉信号无需执行，踢球队员由游戏控制器进行沟通。

3.9.2 执行

必要时，裁判员可能需要放置球。对于推击自由球，球将留在原位，并根据推击规则（见第4.10节）重新定位。如果球离开了场地，球将按照第3.8节所述的方式定位。

在任意球期间，只有进攻队可以接近球，距离不能超过0.75米。防守队的所有机器人必须立即远离球。违反这些规定的防守机器人会受到“非法位置”的处罚（参见第4.4节），该处罚会导致标准移除处罚（参见第4.2节）。在任意球期间对其他机器人（包括推搡）施加的额外处罚不会导致额外的任意球，但仍然使用相应的移除处罚。

如果满足以下条件，自由球将被视为完成，比赛将恢复正常：

- 进攻队明显移动球，但机器人起身除外，该情况不适用此规则。

- The 30 s time period expires (or the game time expires).

The head referee will announce a free kick is completed by calling “Ball Free”, and the GameController resumes the game state `playing`. Note that the sub-state will be automatically left after the 30 s time period expires.

3.10 Penalty Kick

A penalty kick is initiated when a pushing penalty (see Section 4.10) is awarded near the ball against the defending team within their own penalty area. The head referee will announce a penalty kick by calling “Penalty Kick <offending color><offending number>” for the pushing robot.

When the GameController activates the penalty kick, the game changes to the `ready` state with `penalty kick` sub-state active.

This denotes that the robots are given time to set up and prepare for the penalty kick. Similar to kick-off, the game clock is *paused* during finals games only. The referees should pick up the ball.

In Champions Cup only, the GameController does not communicate the identities of the attacking and defending team. Instead, this information is communicated via the same visual gesture described in Section 3.9.1 for free kicks.

Robots have 30 s to get into position for the penalty kick. At the end of 30 s, the game changes to the `set` state with `penalty kick` sub-state active. Similar to a kick-off, during the `set` state the robots are waiting for the penalty kick to commence. Standard penalties apply. Additionally, only the goalkeeper robot from the defending team may be within the penalty area. The goalkeeper robot must either be touching the goal line or stay outside its own penalty area. If the goalkeeper positioned itself inside the goal area but ahead of the goal line it is placed back by the referee during `set` state, so the center of its feet touches the center of the goal line. The placement should happen perpendicular to the goal line without rotating the robot. If this manual placement would lead to a position the robot collides with a goalpost, it is placed inside or outside the goalpost on the goal line, depending on which position is closer. Only one robot from the attacking team may be within the penalty area, and it may not block the penalty mark. Blocking the penalty mark is considered to be an illegal position. All robots that do not reach legal positions will be penalized with the “Illegal Positioning” penalty (see Section 4.4). If multiple robots from the attacking team are in the penalty area in the `set` state, the one robot not to be penalized will be the one that is closest to the penalty mark without blocking it.

The referee should also place the ball on the penalty mark during `set`. The referee signals the penalty kick commences by blowing the whistle once, and calling “Playing”. The game switches to the `penalty kick` sub-state of the `playing` game state, and the game clock is resumed. (Note that the GameController signal is delayed by 15 s when switching from `set` to `penalty kick` sub-state). The attacking team has 30 s to complete the penalty kick.

During the penalty kick:

- 30秒的时间限制到期（或比赛时间结束）。

主裁判将通过呼喊“球自由”来宣布任意球完成，随后游戏控制器将恢复游戏状态。请注意，子状态将在30秒时间期限结束后自动退出。

3.10 罚球

当防守方在己方罚球区内对球门附近的推击犯规（参见第4.10节）时，将启动罚球。主裁判将通过呼喊“罚球 <犯规方><犯规号>”来宣布对推击机器人的罚球。

当游戏控制器激活罚球时，游戏将切换到准备状态，并激活罚球子状态。

这表示机器人将获得时间进行站位和准备罚球。与开球类似，在决赛中游戏时钟才会暂停。裁判应捡起球。

仅在冠军杯中，游戏控制器不会传达攻防双方的身份。相反，这些信息将通过与第3.9.1节中任意球描述的相同视觉手势进行传达。

机器人在30秒内需进入罚球位置。30秒结束后，比赛转为罚球状态，罚球子状态激活。类似于开球，在罚球状态下，机器人等待罚球开始。标准罚球规则适用。此外，只有防守队的守门机器人可以进入罚球区。守门机器人必须接触球门线或停留在己方罚球区外。如果守门机器人位于球门区内但位于球门线前方，裁判会在罚球状态下将其复位，使其脚的中心点与球门线中心点重合。复位应垂直于球门线，不得旋转机器人。如果手动复位会导致机器人与球门柱碰撞，则将其放置在球门线内侧或外侧，以距离更近的位置为准。进攻队只能有一台机器人在罚球区内，且不得阻挡罚球点。阻挡罚球点被视为非法位置。所有未到达合法位置的机器人将被处以“非法站位”罚则（参见第4.4节）。如果罚球状态下进攻队有多台机器人在罚球区内，则不会被罚则的是距离罚球点最近且未阻挡的机器人。

裁判员在开球时也应将球放在罚球点上。裁判员通过吹响哨子一次并喊出“比赛开始”，来示意罚球开始。比赛切换到比赛状态中的罚球子状态，比赛时钟恢复计时。（注意：从开球状态切换到罚球子状态时，GameController信号会延迟15秒）。进攻队在30秒内完成罚球。

在点球阶段：

1. The defensive goalkeeper robot must always be in contact with the goal line if it is inside the penalty area, and must remain on its feet. The goalkeeper is only permitted to “dive”, and be off its feet after the attacking robot has touched the ball.
2. The attacking robot may move freely.
3. All robots must be within the field-of-play. That is, robots may not be outside the field lines, but within the field border.
4. No other robots may enter the penalty area (see Section 4.4).
5. Additional penalties against any further robots, including pushing, do not result in an additional free kick, but still use the appropriate removal penalty.

The attacking robot (taking the penalty kick) may only score a goal if it touches the ball once. Once the attacking robot has touched the ball, it may not score a goal until another robot (from either team) touches the ball. If this robot “scores”, it results in a goal kick (see Section 3.3.1).

The penalty kick is deemed completed if:

- The attacking team touches the ball, even if the robot has fallen.
- The 30 s time period expires (or the game time expires).

The head referee will announce a penalty kick is completed by calling “Ball Free”, and the GameController resumes the game state `playing`. Note that the sub-state will be automatically left (returning to the `playing` state) after the 30 s time period expires.

Note that the restrictions on the attacking robot still apply after the penalty kick is complete.

3.11 Game Stuck

In the event of no substantial change in the game state for 30 s, this is considered a game stuck. “Substantial change” can consist of a robot seeing and moving towards the ball OR robots exploring the field (presumably in an attempt to find the ball).

The main referee has two options how to solve the game stuck and to reestablish the chance of progress in the game. The intention of the game stuck rule is to achieve progress with as little intervention as possible, i.e. the *Local Game Stuck* rule will be preferred, but only if there is a chance that its application will result in progress in the game.

1. 防守门员机器人若在罚球区内，必须始终与球门线接触，并保持站立。门员只能“扑救”，在进攻机器人触球后才能离开地面。
2. 攻击机器人可以自由移动。
3. 所有机器人都必须在比赛区域内。也就是说，机器人不得位于场外，但必须在场边以内。
4. 其他机器人不得进入罚球区（参见第4.4节）。
5. 对任何其他机器人（包括推搡）的进一步处罚，不会导致加罚任意球，但仍然适用相应的移除处罚。任意球，但仍然使用适当的犯规处罚。

进攻机器人（执行罚球）若只触球一次，才能得分。一旦进攻机器人触球，必须等到另一机器人（任何队伍）触球后，才能再次得分。若该机器人“得分”，则判罚球门球（参见第3.3.1节）。

点球被判定完成，如果：

- 进攻队伍触球，即使机器人已经倒下。
- 30秒时间期限结束（或比赛时间结束）。

主裁判将通过喊出“球权恢复”来宣布点球完成，并由游戏控制器恢复游戏状态。请注意，子状态将在30秒时间期限结束后自动离开（返回到比赛状态）。

请注意，在点球完成后，对进攻机器人的限制仍然适用。

3.11 游戏卡住

若游戏状态在30秒内未发生实质性变化，则视为游戏卡死。“实质性变化”可以包括机器人看到并朝球移动，或机器人探索场地（推测是在尝试寻找球）。

主裁判有两种选择来解决游戏卡住的问题，并重新建立游戏进度的机会。游戏卡住规则的目的是以最少干预来实现进度，即优先使用本地游戏卡住规则，但只有在其应用有可能导致游戏进度的情况下才会使用。

3.11.1 Local Game Stuck

If one robot is preventing the game from proceeding—perhaps by circling the ball repeatedly without kicking the ball—it is recommended to improve progress by removing this one robot. The head referee calls “Local Game Stuck <robot>” for this robot, which is penalized (see Section 4.7).

3.11.2 Global Game Stuck

If no robots have made progress towards the ball or began to explore the field in 30 s Global Game Stuck may be called. The referee calls “Global Game Stuck” to announce the decision.

Once the referee calls Global Game Stuck, players enter the `ready` state, and a new kick-off is awarded. Since no team has now kick-off the drop-ball rule should apply, see Section 3.17.

A global game stuck can only be called if at least one robot has touched the ball since the previous kick-off.

3.12 Request for Pick-up

Either team may request that one of their players be picked up (called “Request for Pick-up”). In the `playing` or `ready` state, players may only be picked up for hardware failures. In all other states, players may be picked up for any reason.

Every change (hardware or software) is allowed during a request for pick-up. In particular, it is permitted to change batteries, fix mechanical problems, reboot the robots, and change configuration files. It is also allowed to replace a broken robot by a substitute robot. Substitute robots must wear the correct jersey color for their role (goalkeeper or field player). It is discouraged to change the robot’s control program, **but not forbidden**.

Any strategic “Request for Pick-up” is not allowed. That is, gaining an advantage by removing the robot from the game. In this case, the head referee will indicate when the robot is no longer affecting play and can be removed from the field by an assistant referee.

To prevent mistakes and confusion during games, only team captains (see Section 5.4) should make a “Request for Pick-up”, and only one designated person per team shall accept the robot from the referee, and hand it back after fixing the problem. The returning robot may be returned following the normal replacement procedure once at least 45 s have elapsed since the robot was removed from play. Note that this penalty does not follow the standard removal procedure, and hence does not count towards the incremental penalty count. If the picked-up robot was penalized, the penalty time of the robot counts down with the game clock throughout the pick-up.

The robot should be returned to the assistant referees in the `penalized` state. Note here, that the returning robot or the substitute robot will have to wait out any remaining penalty time of the picked up robot after the team handed their robot back to the assistant referees.

3.11.1 本地卡住

如果一个机器人阻止了游戏的进行——例如通过反复围绕球而不踢球——建议通过移除这个机器人来提高进度。主裁判会对此机器人呼叫“本地游戏卡住 <机器人>”，该机器人将被处罚（见第4.7节）。

3.11.2 全局卡住

如果30秒内没有机器人向球前进或开始探索场地，可能会判定为全局卡住。裁判会宣判“全局卡住”以宣布该决定。

一旦裁判宣判全局卡住，球员将进入准备状态，并判予一次新的开球。由于现在没有队伍开球，应适用丢球规则，见第3.17节。

只有在至少有一个机器人自上次开球以来触球的情况下，才会判定为全局卡住。

3.12 捡拾请求

任何队伍都可以请求替换其中一名球员（称为“请求替换”）。在比赛或准备状态中，球员只能因硬件故障被替换。在其他所有状态下，球员可以因任何原因被替换。

在请求替换期间，允许进行任何更改（硬件或软件）。特别是，允许更换电池、修复机械问题、重启机器人以及更改配置文件。还允许用备用机器人替换损坏的机器人。备用机器人必须穿着与其角色（守门员或场内球员）相符的正确球衣颜色。不建议更改机器人的控制程序，但并非禁止。

任何具有战略性的“请求替换”均不被允许。也就是说，通过移除机器人来获得比赛优势。在这种情况下，主裁判将在机器人不再影响比赛时指示其可以被助理裁判移出场地。

为防止比赛中出现错误和混乱，只有队长（见第5.4节）应提出“请求替换”，并且每队只能指定一人从裁判手中接收机器人，并在修复问题后将其归还。一旦机器人被移出比赛至少45秒后，可以按照正常替换程序将其归还。请注意，此处罚不遵循标准移出程序，因此不计入增量处罚计数。如果被替换的机器人受到处罚，其处罚时间将随着比赛时钟在整个替换过程中倒计时。

机器人应被返回给助理裁判，并处于犯规状态。请注意，返回的机器人或替换的机器人需要在球队将机器人交还给助理裁判后，等待被捡起的机器人剩余的罚时结束。

3.13 Request for Timeout

Each team can call a **maximum of 1 timeout per game** with a total time of no more than **5 minutes**. During this time, both teams may change robots, change programs, or anything else that can be done within the time allotted. During normal game time, a team may call a timeout at any stoppage of play (after a goal, stuck game, before a half, etc.). Alternatively, a team may call a timeout before a penalty shootout if they have not used their timeout yet (see Section 3.19).

The timeout ends when the team that called the timeout says they are finished, at which time they must be ready to play. The other team must be ready to play at the time the timeout runs out, or **2 minutes** after a prematurely called end of the timeout, whichever is earlier. If the other team is not ready to play in time, it has to call a timeout of its own.

The clock stops during timeouts, even during the preliminaries, and is reset to the time when the current stoppage of play began.

After the completion of the timeout, the game resumes with a kick-off for the team which did not call the timeout.

If a team is not ready to play at the assigned time for a game, the referee will call the timeout for that team. After the expiration of such a timeout, if the team is still not ready to play then the referee shall start the game with only one team on the field. The team that was not ready can return its robots to the field as per the rules for “Request for Pick-up”. If both teams are not ready, the referee will call timeouts for both teams. This “double timeout” expires after 10 minutes.

3.14 Referee Timeout

The head official may call a timeout at any stoppage of play if he or she deems it necessary. A referee timeout should only be called in dire circumstances—one example might be when the power to the wireless router is down or no robot listens to the GameController. However, when and whether to call a referee timeout is left up to the head referee.

Referees may call multiple timeouts during a game if needed. Teams may do anything during these timeouts, but they must be ready to play **2 minutes** after the referee ends a timeout. The referee should end the timeout once he or she believes the circumstance for which the timeout was called has been resolved. In cases where the circumstance for which the timeout was called is not resolved within 10 minutes, the chair of the technical committee should be consulted regarding when/if play should continue.

The team who would have kicked off if the timeout had not been called shall kick-off when the game resumes.

3.13 超时请求

每队每场比赛最多可叫1次暂停，总时长不超过5分钟。在此期间，两队可以更换机器人、更换程序，或做任何在规定时间内可以完成的事情。在正常比赛时间内，球队可以在任何停赛点（进球后、比赛卡住、半场前等）叫暂停。或者，如果球队尚未使用暂停，他们可以在点球大战前叫暂停（见第3.19节）。

当请求超时的队伍表示完成时，超时即结束，此时他们必须准备好继续比赛。另一队必须在超时结束时或提前结束超时后的2分钟内准备好比赛，以较早者为准。如果另一队未能及时准备好比赛，则必须自行请求超时。

在超时期间，计时器会停止，即使在预赛阶段也是如此，并且会重置为当前比赛暂停开始时的时间。

超时结束后，比赛将以未请求超时的队伍开球的方式恢复。

如果队伍在指定的比赛时间未能准备好比赛，裁判将为其请求超时。在该超时结束后，如果该队伍仍然未能准备好比赛，裁判应仅有一队在场的情况下开始比赛。未准备好的队伍可以根据“请求取回”的规则将机器人送回场上。如果两队都未能准备好，裁判将分别为两队请求超时。这种“双超时”在10分钟后到期。

3.14 裁判暂停

主裁判可以在任何比赛暂停时，如果他认为有必要，可以叫暂停。裁判暂停只有在极端情况下才应被叫——例如，当无线路由器断电或没有机器人响应游戏控制器时。然而，何时以及是否叫裁判暂停，由主裁判决定。

裁判在比赛中可以根据需要叫多次暂停。在暂停期间，队伍可以做任何事，但在裁判结束暂停后，他们必须在2分钟内准备好继续比赛。裁判应在他认为暂停的原因已经解决时结束暂停。如果暂停的原因在10分钟内仍未解决，应咨询技术委员会主席，商讨何时/是否应继续比赛。

如果计时请求未被调用，本应开球的队伍将在比赛恢复时开球。

3.15 Extra Time

The head official may decide to add time to the clock if a substantial delay (such as an enormous wireless delay) causes excessive game time to be lost. The decision to add time to the clock should be made immediately after the incident. The GameController operator should execute this addition of time using the GameController interface.

3.16 Mercy Rule

A game will conclude once the game score shows a goal difference of 10. Ending the game is mandatory once a goal difference of 10 is reached.

3.17 Drop-Ball Rule

In the event that there has been a referee error or a pause in game during the `playing` state and it has been acknowledged by the head referee, the drop-ball rule shall apply. The game must stop and will start again when the arising issue has been resolved. This rule also applies for the Global Game Stuck, see Section 3.11.2.

When resuming the game, the process should be the same as an initial kickoff as specified in Section 3.6, placing the robots on the sidelines. However, none of the teams will be given kickoff, and no team will be allowed in the center circle until the `playing` state, as there is no longer time allotted for kick-off. With the transition into `playing` the ball is directly in play.

The only current situations where the drop-ball rule would be required is when a referee timeout or a global game stuck has been called.

3.18 Ball Stop Rule

If a game half were to extend longer than the allotted time, due to an attack situation by a team, the half should end if the ball comes to a complete stop or when the ball leaves the field, with no immediate play by other robots.

3.19 Penalty Kick Shoot-Out

A penalty kick shoot-out is used to determine the outcome of a tied game when an outcome is required (for example, when team progression is tied on all tie-break factors, during the promotion round, intermediate round, quarterfinals, semifinals, third place or final).

3.15 加时赛

当出现重大延误（例如巨大的无线延迟）导致比赛时间严重损失时，主裁判可以决定为计时器增加时间。增加计时器时间的决定应在事件发生后立即做出。GameController操作员应使用GameController界面执行此时间增加操作。

3.16 宽容规则

当比分出现10个进球差距时，比赛将结束。一旦达到10个进球差距，必须结束比赛。

3.17 掷球重新开始规则

如果裁判出现错误或比赛在比赛状态中暂停，并且由主裁判确认，则适用掷球重新开始规则。比赛必须停止，并在问题解决后重新开始。此规则也适用于全球比赛卡住，见第3.11.2节。

重新开始比赛时，过程应与第3.6节中规定的初始开球相同，将机器人放置在边线上。但是，两队都不会获得开球机会，直到进入比赛状态，因为开球没有更长的时间限制。随着球的过渡，球直接进入比赛状态。

只有在裁判暂停或全局游戏卡顿时，才需要使用落球规则。

3.18 球停止规则

如果由于某队进攻情况，半场时间需要延长超过规定时间，当球完全停止或离开场地且其他机器人没有立即进行比赛时，该半场应结束。

3.19 罚球点球大战

点球大战用于决定平局比赛的胜负，当需要确定结果时（例如，当球队在所有平分因素上都相同，在升级轮、中间轮、四分之一决赛、半决赛、季军赛或决赛中）。

All penalty kicks are taken against the same goal.⁴ The team listed first on the competition schedule will have the striker robot for the first penalty kick. Subsequently, both teams take kicks alternately. At all stages of the competition, the penalty kick shoot-out will consist of three penalty kicks per team. A team that has scored the most goals at the conclusion of these will be declared the winner. A winner can also be declared before the conclusion of the penalty shoot-out if a team can no longer win. If the two teams remain tied after three penalty kicks, then a sudden death shoot-out will follow until a definite winner is found.

The penalty kick shoot-out starts immediately without changing any code after the second half ended. No timeouts may be called during the penalty shoot-out. However, a team may request a timeout before the penalty shoot-out starts if they have a timeout remaining for this game. During the timeout code changes are allowed.

Before the penalty shoot-out begins, each team must hand over to referees up to 6 prepared robots that may participate in the penalty shoot-out. No robots may be added once the penalty shoot-out starts. Robots that will not participate in the shoot-out must not be on the wireless network and must stay outside of the field. All participating robots must be wearing the correct jersey for their player and no duplicate numbers are permitted.

During the entire penalty kick shoot-out, the robots are controlled by the GameController and the referee's whistle. A team cannot request the referees to press buttons on their robots, except for initially leaving the `unstiff` state.

3.19.1 Penalty Kick

A penalty kick is carried out with one striker robot and one opposing goalkeeper. The penalty kick commences with the `set` game state activated. Before each penalty kick, both teams must select the robot to participate (as goalkeeper or striker) in the penalty kick. The team leader communicates the selection to the head referee by privately handing the referee a card with their chosen number. After both teams have selected their player, the GameController operator selects the requested striker and goalkeeper robots from the opposing teams and the GameController communicates that all non-selected robots are substitutes and should remain inactive. The GameController indicates which team has the striker robot in the current penalty kick.

Referees place the ball, the striker, and goalkeeper robots. The ball is placed on the penalty mark closest to the goal being defended. The striker robot is positioned on the edge of the penalty area, facing the ball and the goal. The goalkeeper is placed with its feet on the goal line and in the center of the goal. Neither robot is permitted to locomote (move their legs) during the `set` state. Movement of the robot's head and arms is allowed.

The head referee commences the penalty kick by blowing the whistle *once*, and calling "Playing". The GameController activates the penalty kick, switching to the `playing` game state. Note that the `playing` signal is delayed (see Section 3.2).

⁴Which goal to take for the shoot-out is decided in accordance with the teams, or otherwise by a coin toss.

所有点球都对同一个球门进行。⁴ 比赛日程表上列出的第一支球队将获得第一个点球的权利。之后，两队交替进行点球。在整个比赛过程中，点球大战将包括每队三个点球。在完成这些点球后，进球最多的队伍将被宣布为胜者。如果在点球大战结束前，有一队已经无法获胜，也可以宣布胜者。如果两队在三轮点球后仍然平分，那么将进入加时赛，直到找到明确的胜者。

加时赛点球大战在下半场结束后立即开始，且无需更改任何代码。点球大战期间不得请求暂停。不过，如果队伍在本场比赛中还有剩余暂停次数，可以在点球大战开始前请求暂停。暂停期间允许进行代码更改。

点球大战开始前，每支队伍必须向裁判提交最多6台准备好的机器人参与点球大战。点球大战开始后不得添加任何机器人。不参与比赛的机器人不得连接无线网络，且必须停留在场外。所有参与比赛的机器人必须穿着对应球员的正确球衣，且号码不得重复。

在整个点球大战期间，机器人由游戏控制器和裁判的哨声控制。队伍不得请求裁判按下机器人上的按钮，除非是初始解除僵硬状态。

3.19.1 点球

点球由一名前锋机器人和一名对方守门机器人进行。点球开始时激活预设的游戏状态。每次点球前，两队必须选择参与点球（担任守门员或前锋）的机器人。队长通过私下将写有选定号码的卡片交给裁判的方式，向主裁判传达选择。两队选定球员后，游戏控制器操作员从对方队伍中选定请求的前锋和守门机器人，游戏控制器通知所有未选中的机器人是替补且应保持非活动状态。游戏控制器指示当前点球中拥有前锋机器人的队伍。

裁判员放置球、前锋机器人和守门员机器人。球被放置在距离防守方球门最近的罚球点上。前锋机器人位于罚球区边缘，面向球和球门。守门员被放置在球门线上，位于球门中央。在准备状态下，两个机器人都允许移动（用腿移动）。机器人的头部和手臂可以移动。

主裁判通过一声哨响并喊出“比赛开始”来开始罚球。游戏控制器激活罚球，切换至比赛状态。请注意，比赛信号有延迟（参见第3.2节）。

⁴点球大战选择哪个球门，由双方球队决定，或者通过抛硬币决定。

The time limit for the striker is 30 s after the penalty kick starts. A penalty kick is over when the ball has come to a full stop after the first contact by the striker robot. The striker robot is not allowed to play the ball a second time after the ball has clearly moved, otherwise the penalty kick ends immediately as well. A goal is awarded to the attacking team if a goal has been scored (i. e. the ball has completely crossed the goal line) before the penalty kick is over. Otherwise, the score is unchanged.

The goalkeeper robot must always be in contact with the goal line and must remain on its feet until the striker robot touches the ball. The goalkeeper is only permitted to “dive” and be off its feet after the attacking robot has touched the ball. If the goalkeeper violates these rules, then the penalty kick ends immediately and a goal will be awarded to the attacking team.

All rules such as “Ball Holding”, “Pushing” and others are applied during the penalty kick. A goalkeeper will not be penalized for inactivity during a penalty kick, provided its stiffness is on. Other penalties are applied as usual.

3.19.2 Sudden Death Shoot-Out

Teams take one additional penalty kick each, and the game decision will be made as follows:

- Each kick is ranked in one of the following categories:
 1. goal
 2. kick at the goal that is blocked by the goalkeeper
 3. kick that hits a goalpost
 4. kick that crosses the goal line next to the goalposts
 5. any other kick
 6. no kick
- If both teams’ kicks rank in different categories, the team with the higher ranked kick wins.
- Otherwise, the sudden death is repeated.
- If after 3 sudden death penalty kicks there is still no winner, the referee will toss a coin to decide the game.

进攻方球员在罚球开始后有30秒的时间限制。罚球在进攻方机器人首次触球后，球完全静止时结束。球明显移动后，进攻方机器人不得再次触球，否则罚球也将立即结束。如果在罚球结束前进球（即球完全越过球门线），则判进攻方得分。否则，比分保持不变。

守门员机器人必须始终与球门线接触，并且必须保持站立状态，直到进攻机器人触球。守门员只有在进攻机器人触球后才能“扑救”并失去平衡。如果守门员违反这些规则，则点球将立即结束，并且进球将判给进攻队。

“控球”、“推搡”等所有规则在点球阶段均适用。只要守门员的刚性开启，其在点球阶段的无活动行为不会受到处罚。其他处罚按常规执行。

3.19.2 加时突然死亡赛

双方球队各进行一次额外的点球，比赛结果将按以下方式判定：

- 每次射门将被归类到以下类别之一：
 1. 进球
 2. 踢向被守门员挡住的球门
 3. 踢中门柱的球
 4. 踢过门柱附近球门线的球
 5. 其他任何踢球
 6. 没有踢球
- 如果两队的踢球属于不同类别，那么排名较高的队伍获胜。
- 否则，突然死亡会重复发生。
- 如果在3次突然死亡判罚后仍然没有胜者，裁判将抛硬币决定比赛结果。

4 Forbidden Actions and Penalties

The following actions are forbidden. In general, when a penalty applies, the robot shall be replaced, not the ball.

4.1 Penalty Procedure

When a robot commits an infraction, the head referee shall call out the **infraction** committed, the **primary jersey color** of the robot, and the **jersey number** of the robot. The penalty for the infraction will be applied immediately by an assistant referee. The assistant referees should perform the actual movement of the robots for the penalty so that the head referee can continue focusing on the game. The operator of the GameController will send the appropriate signal to the robots indicating the infraction committed.

For penalties that are timed, the penalty time is considered to be over at the end of each half.

4.2 Standard Removal Penalty

Unless otherwise stated, all infractions result in the removal of the infringing robot from the field of play for a particular amount of time, after which it will be returned to the field of play. This process is called the *standard removal penalty*.

When the head referee indicates an infraction has been committed that results in the standard removal penalty, the assistant referee closest to the robot will remove the robot immediately from the field of play. The robot should be removed in such a way as to minimize the movement of the other robots and the ball. If the ball is inadvertently moved when removing the robot, the ball should be replaced to the position it was in when the robot was removed.

The GameController will send the appropriate penalty signal to the robot indicating the infraction committed. After a penalty is signaled to the robot, it is not allowed to move in any fashion (except for standing up).

The initial duration of the standard-removal-penalty-time is 45 s. Unless otherwise specified, the penalty time increases by 10 s each time a team commits any infraction. During the `set` state the penalty time counter will not decrease. The GameController will keep track of the time of the penalty.

The removed robot will be placed outside the field at a distance of approximately 50 cm away from the nearest touchline, facing **towards** the field of play. It should be placed close to the position where the penalty point projects on the touchline. This is illustrated in Figure 10. If there is another robot already in this position, the robot should be placed at a nearby location along the touchline.

4 限制行为与处罚

以下行为禁止。通常，当处罚适用时，应更换机器人，而不是球。

4.1 处罚程序

当机器人犯规时，主裁判应报出犯规行为、机器人主要球衣颜色以及机器人球衣号码。助理裁判应立即执行该犯规的处罚。助理裁判应负责执行机器人实际移动，以便主裁判能继续专注于比赛。游戏控制器操作员将向机器人发送适当信号，表明犯规行为。

对于有时间限制的处罚，处罚时间在每个半场结束时视为结束。

4.2 标准移除处罚

除非另有说明，所有犯规行为都将导致犯规机器人被移出比赛场地一段时间，之后将被送回比赛场地。这个过程称为标准移除处罚。

当主裁判示意犯规导致执行标准罚球时，距离机器人最近的助理裁判将立即将机器人移出比赛场地。移除机器人时应尽量减少其他机器人和球体的移动。若在移除机器人时不慎移动了球，应将球放回机器人被移除时的位置。

游戏控制器将向机器人发送相应的罚球信号，表明犯规行为。一旦对机器人发出罚球信号，机器人不得以任何方式移动（除站立外）。

标准罚球时间的初始时长为45秒。除非另有规定，每当队伍犯规时，罚球时间会增加10秒。在设定状态下，罚球时间计数器不会减少。游戏控制器将记录罚球时间。

被移除的机器人将被放置在场外，距离最近边线约50厘米，面向比赛场地。它应放置在罚球点投影在边线上的位置附近。这如图10所示。如果该位置已有另一台机器人，则应将这台机器人放置在边线附近的其他位置。

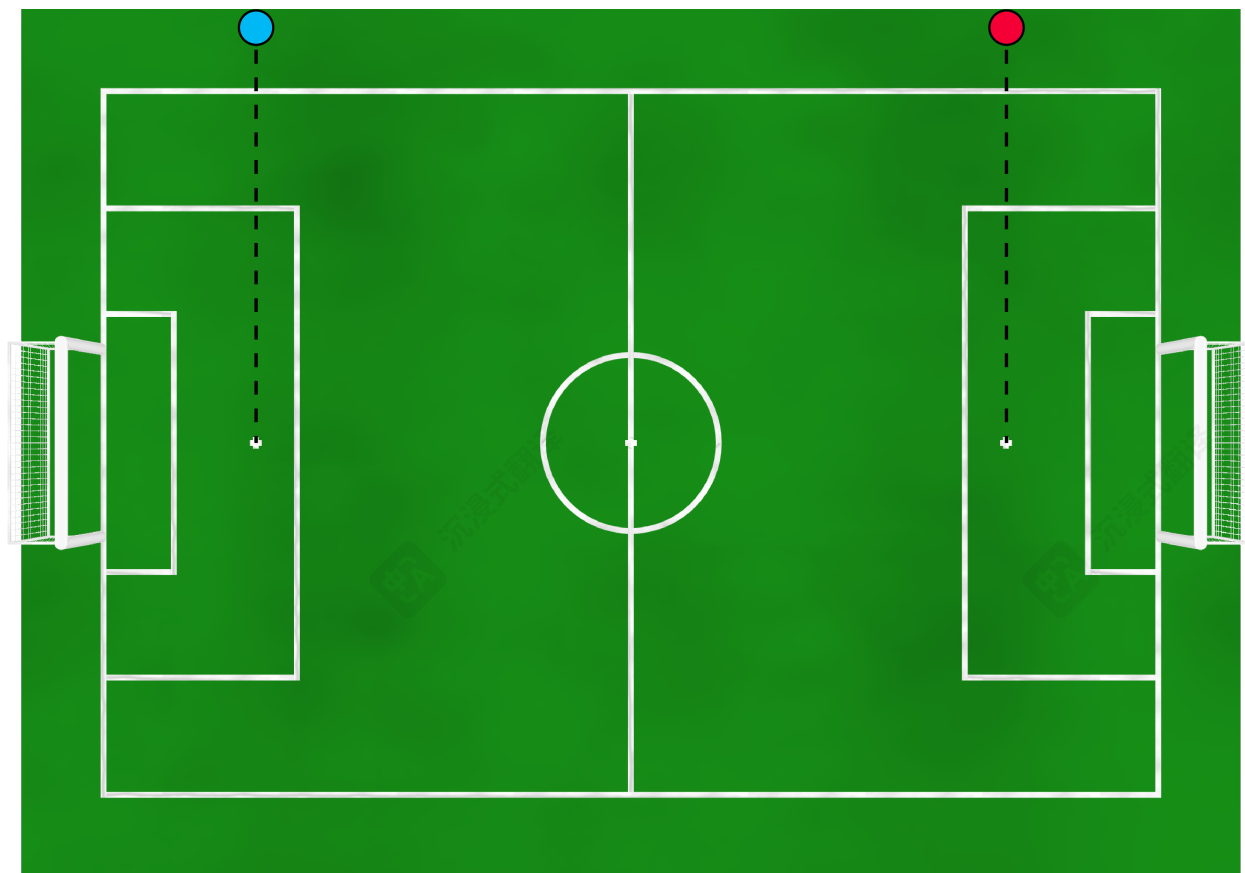


Figure 10: For robots coming back from a standard removal penalty, re-entry points are in line with or around the penalty mark in their own half, but always where the robots were placed at the beginning of the penalty.

When finding a nearby location it **must** still be in the robot's own half, so that the symmetry of the field can be resolved by the robot's localization system.

After the penalty time elapses, the GameController operator will remove the penalty from the penalized robot, enabling it to autonomously re-enter the game without requiring additional intervention from the referees.

4.3 Forbidden Actions

The following actions are forbidden, but not treated as penalties. Each forbidden action specifies the actions to be taken by the referees.

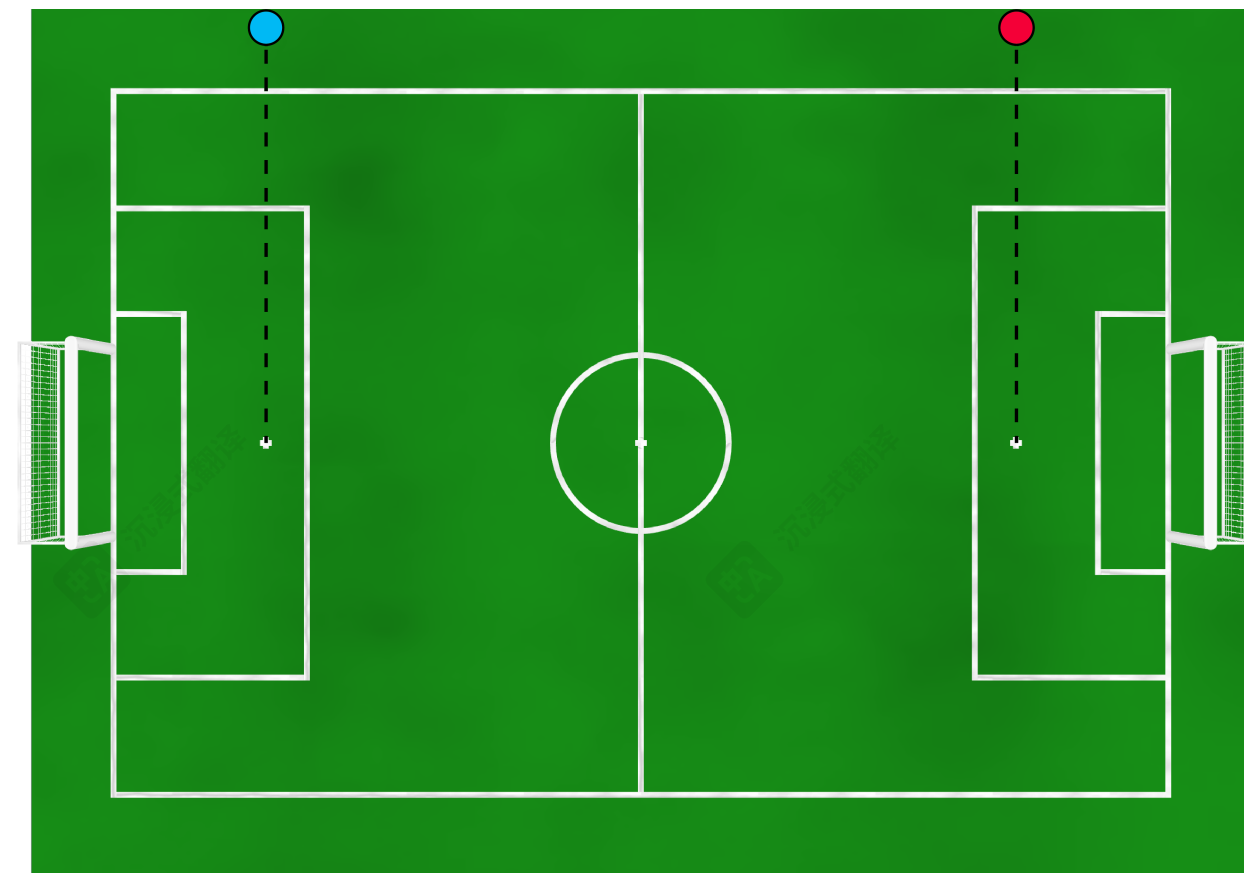


图10：对于从标准移除罚中返回的机器人，重新进入点与其所在半场的罚球点对齐或在其附近，但始终是罚的开始时机器人放置的位置。

在寻找附近位置时，它必须仍在机器人自己的半场内，以便机器人的定位系统能够解决场地的对称性问题。

在罚时结束后，游戏控制器操作员将从受罚机器人中移除罚，使其能够自主重新进入比赛，而无需裁判进行额外干预。

4.3 禁止行为

以下行为被禁止，但不视为处罚。每项禁止行为都规定了裁判员应采取的行动。

4.3.1 Manual Interaction by Team Members

Manual interaction with the robots, either directly or via some communications mechanism, is not permitted. Team members can only touch one of their robots when an assistant referee hands it over to them after a “Request for Pick-up”.

4.3.2 Damage to the Field

A robot that damages the field, or poses a threat to spectator safety, will be removed from the field for the remainder of the game.

4.4 Illegal Positioning

A robot penalized under illegal position has the “Illegal Position” penalty applied. Illegally positioned robots are subject to the standard removal penalty (see Section 4.2). The head referee will call “Illegal Position <robot>”. Referees may interchange “Illegal Position” with “Illegal Defender” to help with clarity.

Illegal positions are described below.

For simplicity, Illegal Positioning penalties during the `set` state (for kick-off or a penalty kick) do not count towards the incremental penalty count.

Refer to Section 1.6 for the definition of *inside/outside* of a region of the field.

4.4.1 Before and During Kick-off

If a robot violates the positioning restrictions made in Section 3.7 at the time the `set` state starts, it will be penalized and removed for 15 s. If a robot enters the areas of the other team before the ball is in play, it will be penalized and removed according to the standard removal penalty. A robot making reasonable effort to return to the field from a standard penalty must not be penalized for illegal position.

4.4.2 Own Goal Area

During the `set` and `playing` states, at most *three* players can be within their own goal area at the same time.

4.3.1 成员手动干预

禁止直接或通过某种通信机制与机器人进行手动交互。在助理裁判员在“请求取走”后递给他们机器人时，队员只能触摸自己的一个机器人。

4.3.2 对赛场的损害

对赛场造成损害的机器人，或对观众安全构成威胁的机器人，将在剩余比赛中被移出赛场。

4.4 非法位置

受到非法位置处罚的机器人将被施加“非法位置”处罚。处于非法位置的机器人将受到标准移除处罚（见第4.2节）。主裁判将呼叫“非法位置 <机器人>”。裁判员可以将“非法位置”与“非法防守者”互换，以帮助提高清晰度。

非法位置描述如下。

为简化起见，在此状态下（例如开球或罚球）的非法定位处罚不计入增量处罚次数。

有关场地区域内/外的定义，请参阅第1.6节。

4.4.1 开球前及开球时

如果机器人在设定状态开始时违反了第3.7节中规定的定位限制，将被处罚并移除15秒。如果机器人在球未进入比赛状态前进入对方队伍的区域，将被根据标准移除处罚进行处罚并移除。机器人从标准处罚中做出合理努力返回场地的行为不应因非法位置而受到处罚。

4.4.2 球门区

在设置和比赛状态下，最多三名球员可以同时位于自己的球门区域内。

A robot is within the goal area if any part of its body is touching the ground inside the goal area or touching one of its lines. The penalty is applied when any additional players enter the area in `playing`, or to the excessive players closest to the border of the goal area in `set`. Note that if a player is pushed into the goal area by an opponent, this robot will not be subject to removal, unless it fails to exit the area within 5 s (or 5 s after getting up if the pushing led to falling).

If an illegal defender kicks an own goal, the goal is scored for the opponent. If there is any doubt about whether a goal should count (e. g. the illegal defender infraction is called, but the robot scores the own goal immediately afterwards, before it is removed) then the decision shall be against the infringing robot.

4.4.3 Defender Encroachment During Free Kick

If a robot of the defensive team enters or does not attempt to leave the circular area with 0.75 m radius around the ball after a free kick (see Section 3.9) was called, “Illegal Position” is called. Note that the referee should not look for exact distances and rather penalize only those robots who clearly violate this rule. As a guideline, the robots of the defensive team should clear the ball within 10 s.

4.4.4 Penalty Area During Penalty Kick

If a robot enters the relevant penalty area during a penalty kick, except for the goalkeeper (defending team) and one robot of the attacking team, “Illegal Position” is called.

4.5 Forbidden Motion

Robots that begin moving their legs or locomote in any fashion before reaching the specified state in the following subsections will be penalized *in place* on the field. “Forbidden Motion” penalties do not follow the standard removal procedure, and hence do not count towards the incremental penalty count. A robot will be moved back to its original position if it has moved significantly before being penalized.

4.5.1 Motion in Standby

If a robot begins to move during `standby` (i. e. before the head referee does the visual signal), all robots from that team will be penalized from that moment until the delayed *ready* signal is sent. Specifically, for the Champion Cup, this penalty will last in *ready* up to 45 s, while for the Challenge Shield, it will extend up to 35 s. However, movement of the head and initially standing up is permitted and does not result in a penalty. The head referee will call “Forbidden Motion in Standby<team>”. Any response to a false signal of any kind will result in this penalty.

如果机器人的任何部分接触到球门区域内的地面或其线条，则该机器人位于球门区域内。在比赛状态下，当有其他球员进入该区域时将施加处罚，或在设置状态下，对最靠近球门区域边界的多余球员施加处罚。请注意，如果一名球员被对手推入球门区域，该机器人不会被移除，除非它在5秒内（或在被推倒后5秒内站起来）未能退出该区域。

如果非法防守者踢了自球，进球将判给对方。如果对进球是否有效有任何疑问（例如，非法防守者犯规被判罚，但机器人立即踢了自球，且在机器人被移除之前），则裁决应针对犯规的机器人。

4.4.3 防守者越位 界外球

如果防守队的机器人进入或未试图离开自由球（见第3.9节）后，在以球为中心、半径0.75米的圆形区域内，‘非法位置’将被判罚。请注意，裁判不应寻找精确距离，而应仅处罚那些明显违反此规则的机器人。作为参考，防守队的机器人应在10秒内将球清出该区域。

4.4.4 犯规区 罚球时

在罚球时，如果除守门员（防守队）和进攻队的一名机器人外，有机器人进入相关犯规区，将被判罚‘非法位置’。

4.5 禁止动作

在以下小节中指定的状态下，如果在开始以任何方式移动其腿部或移动装置之前，机器人将被当场处罚。‘禁止动作’处罚不遵循标准的移除程序，因此不计入增量处罚次数。如果机器人在被处罚前已经显著移动，它将被移回原位。

4.5.1 待机状态下的动作

若机器人在待机期间开始移动（即裁判长发出视觉信号之前），则该队伍的所有机器人从那时起将被判罚，直到延迟的待机信号被发出。具体来说，对于冠军杯，该判罚将在待机阶段持续长达45秒，而对于挑战盾，则将延长至35秒。然而，头部移动和初始站立是被允许的，不会导致判罚。裁判长将喊出“待机期间禁止移动<队伍>”。对任何类型的错误信号做出反应都将导致此判罚。

4.5.2 Motion in Set

If a robot begins moving during `set` (i.e. before the head referee blows the whistle), it will be penalized for 15 s in the *playing* state. The head referee will call “Forbidden Motion in Set <robot>”. Note that falsely responding to a whistle on another field will result in this penalty.

4.6 Fallen or Inactive Robots

If a robot falls during the game, it should start executing a get-up action within 5 s. If it does not commence a get-up action within 5 s, it will be penalized and removed for 45 s. A fallen robot which is unable to autonomously stand up within 2 attempts, or 3 if disturbed externally during the first two attempts, will be penalized and removed for 45 s. In both cases, the head referee will call “Fallen Robot <robot>”. The goalkeeper, inside its own penalty area, is the only robot permitted to ‘dive’ (that is deliberately falling in a way that might cause its torso, arms, or hands) to intercept the ball. In all other cases, the robot should remain upright—that is, supported by its feet.

A robot that has ceased activity for 10 s or has turned off will be removed and penalized for 45 s. The head referee will call “Inactive Robot <robot>”. A robot is active if it performs at least one of the following:

1. The robot walks in any direction, or turns.
2. The robot searches for the ball, or is looking at the ball.

Fallen/Inactive Robot penalties do not follow the standard removal procedure, and hence do not count towards the incremental penalty count.

4.6.0.1 Note: The intention of this rule is not to penalize robots simply for being stationary—provided they are not ‘asleep’ and have not ‘crashed’.

4.7 Local Game Stuck

When Local Game Stuck is called, the nearest robot to the ball will be penalized and removed for 45 s. Local Game Stuck penalties do not follow the standard removal procedure, and hence do not count towards the incremental penalty count.

4.5.2 集合中的运动

若机器人在比赛准备期间开始移动（即裁判长吹响哨子之前），它将在比赛状态下被罚15秒。裁判长将喊出“比赛准备期间禁止移动<机器人>”。请注意，对其他场地上哨声的错误反应也将导致此判罚。

4.6 倒地或非活动机器人

如果机器人在比赛过程中倒下，应在5秒内开始执行起身动作。如果在5秒内未开始起身动作，将被判罚并移除45秒。如果一个倒地的机器人在2次尝试内无法自主站起来，或者在第一次尝试时受到外部干扰而无法在后续尝试中站起来（最多3次尝试），将被判罚并移除45秒。在这两种情况下，主裁判将呼叫“倒地机器人 <机器人>”。守门员在其自己的罚区内是唯一被允许进行‘跳水’（即故意以可能使其躯干、手臂或手倒下的方式）来拦截球的机器人。在所有其他情况下，机器人应保持直立——即由其双脚支撑。

一个停止活动10秒或已关闭的机器人将被移除并判罚45秒。主裁判将呼叫“非活动机器人 <机器人>”。如果机器人执行以下至少一项操作，则视为活跃：

1. 机器人可以向任何方向行走，或转向。
2. 机器人搜索球，或正在看球。

3. 倒下/非活动机器人罚分不遵循标准移除程序，因此不计入增量罚分计数。

4.6.0.1 注意：本规则的意图并非单纯因机器人静止而罚分——只要它们不是“睡眠”状态且未“崩溃”。

4.7 本地游戏卡住

当调用“本地游戏卡住”时，离球最近的机器人将被处罚并移除45秒。本地游戏卡住的处罚不遵循标准移除程序，因此不计入增量处罚次数。

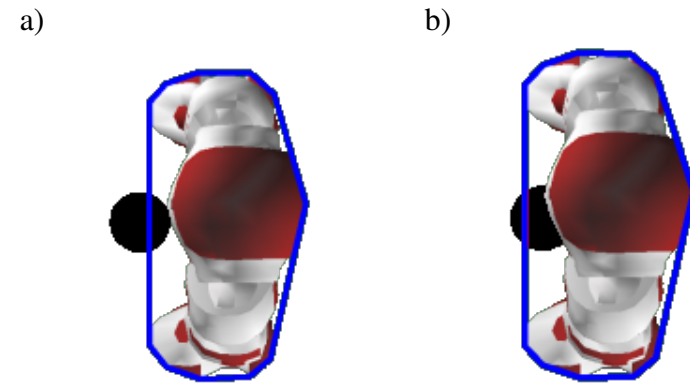


Figure 11: Examples for “Ball Holding”. The black circle is the ball and the blue polygon visualizes the convex hull of the robot’s projection onto the ground. Situation a) is legal, whereas b) violates the rule.

4.8 Ball Holding

The goalkeeper is allowed to hold the ball for up to 10 s as long as it has one foot inside in its own penalty area. In all other cases (except those noted in Section 4.8.1), robots are allowed to hold the ball for up to 3 s. Holding the ball for longer than this is not allowed. The head referee will call “Ball Holding <robot>”, and the robot removed under the standard removal penalty. The ball should be removed from the possession of the robot and placed where the penalty occurred. If the robot that held the ball has moved the ball before the robot can be removed, the ball shall be replaced where the penalty occurred. This applies to accidental goals.

Example. A robot holds the ball, and before the referees can remove the robot, it shoots the ball into the goal. The goal will not be counted, and the ball will be replaced where the penalty occurred.

A robot must leave enough open space around the ball. The occupation of the ball is judged using the convex hull of the projection of the robot’s body onto the ground. “Enough open space” means that at least the half of the ball is not covered by the convex hull. It is not important whether the robot actually touches the ball.

Intentional continual holding is prohibited even if each individual holding time does not continue for up to the time limit. In general, robots should release the ball for approximately as long as they were holding it to reset the clock. Without a sufficient release, the continual holding is regarded as a continuous hold from the very beginning and the holding rule is strictly applied.

4.8.1 Exceptions to the Ball Holding Rules

The following define situations where ball holding does not apply:

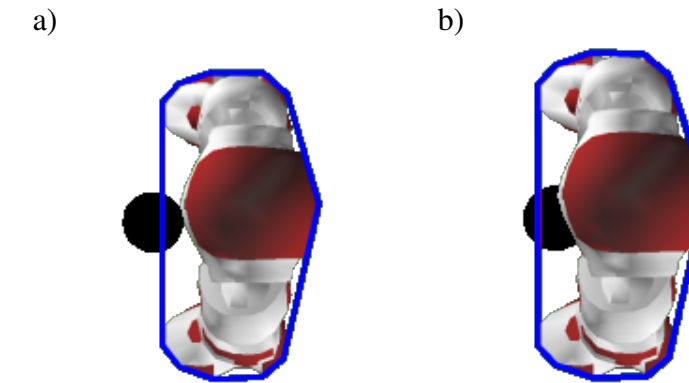


图11：“持球”示例。黑色圆圈是球，蓝色多边形可视化机器人投影到地面上的凸包。情况a)是合法的，而b)违反了规则。

4.8 持球

守门员在自身罚区内只要有一只脚在里面，可以持有球长达10秒。在所有其他情况下（除第4.8.1节中注明的情况外），机器人可以持有球长达3秒。持有球超过这个时间是不允许的。主裁判将呼叫“球持 <机器人>”，并按照标准移除处罚移除该机器人。球应从持有机器人的控制中移除，并放置在处罚发生的位置。如果持有球的机器人在被移除前已经移动了球，球应替换到处罚发生的位置。这适用于意外进球。

示例。机器人持球，在裁判员移开机器人之前，它将球射入球门。球门不会计数，球将被放置在犯规发生的位置。

机器人必须在球周围留出足够的空间。球的占据情况通过机器人身体投影到地面上的凸包来判断。“足够的空间”意味着球的至少一半没有被凸包覆盖。机器人是否实际接触球并不重要。

即使每个单独的持球时间不超过时间限制，故意持续持球也是被禁止的。通常，机器人应该释放球的时间大约等于他们持球的时间来重置计时器。如果没有充分的释放，持续持球被认为从一开始就是连续持球，并且严格应用持球规则。

4.8.1 棒球持球规则的例外

以下定义了球控不适用的情况：

1. Ball holding may not occur when the ball becomes stuck between a robot's legs. In such a situation, the head referee should call 'clear ball' and an assistant referee should remove the ball and place the ball approximately where it was before it became stuck.
2. Ball holding may not occur when a robot falls on a ball. The robot will either get up and hence free the ball, or the robot should be removed under the Fallen Robot rule.

4.9 Player Stance

Robots must not maintain a stance wider than the width of their shoulders by more than one robot hand width on each side for longer than 10 s. This duration should be adequate for a robot gesture. Staying in a wide stance for longer than 10 s will result in the standard removal penalty.

This rule applies to upright robots only, i. e. robots that are supported by their feet. Other cases are handled by the fallen robot rule (see Section 4.6).

4.10 Player Pushing

Pushing is a direct or indirect forceful contact with any other opponent robot, i. e., enough to destabilize it, and is not allowed. In the following, exceptions to this rule are specified in more detail. The head referee will call "Pushing <robot>".

If the ball moves significantly as the result of pushing, then it should be replaced to where it was at the time of the infraction.

A **Pushing Free Kick**, see Section 3.9, is awarded against the robot penalized for pushing

1. if the robot is in an approximately 0.5 m radius around the ball and
2. if the pushed robot was **not** inside the penalty area of the pushing robot.

A **Penalty Kick**, see Section 3.10, is awarded against the robot penalized for pushing

1. if the robot is in an approximately 0.5 m radius around the ball and
2. if the pushed robot was inside the penalty area of the pushing robot.

The following exceptions define situations where pushing does not apply:

- Pushing may occur **only** between players of different teams.

1. 当球卡在机器人腿之间时，可能无法进行球持有。在这种情况下，主裁判应鸣哨‘清球’，助理裁判应将球移开，并将其放置在球卡住之前的大致位置。
2. 当机器人掉落在球上时，球控可能不会发生。机器人会站起来从而释放球，或者机器人应按照倒下机器人规则被移除。

4.9 球员姿态

机器人的姿态宽度不得超出其肩宽，每侧超出机器人一只手宽度，持续时间不得超过10秒。这个时长应足以完成一个机器人手势。如果宽度姿态持续超过10秒，将导致标准移除处罚。

本规则仅适用于直立机器人，即由双脚支撑的机器人。其他情况由倒地机器人规则处理（见第4.6节）。

4.10 球员推搡

推搡是指与其他任何对手机器人直接或间接的强力接触，即足以使其失去平衡，这是不被允许的。在下文中，将更详细地说明此规则的例外情况。主裁判将鸣哨‘推搡 <机器人>’。

如果球因推撞而移动显著，则应将其替换到违规发生时所在的位置。

推撞任意球，参见第3.9节，将判给因推撞被罚的机器人。

1. 如果机器人在球周围约0.5米的范围内，
2. 如果被推的机器人不在推撞机器人的罚区内。

罚球，参见第3.10节，将判给因推撞被罚的机器人。

1. 如果机器人在球周围约0.5米的范围内，
2. 如果被推动的机器人位于推动机器人的罚区内。

以下例外定义了推动不适用的情况：

- 推进只能发生在不同队伍的玩家之间。

- A stationary robot cannot be penalized for pushing, including a robot that is kicking, provided that the ball was close enough where a kick could have succeeded at the start of the kick motion.
- A robot currently getting up cannot be penalized for pushing.
- The goalkeeper cannot be penalized for pushing while looking at or chasing the ball in its own penalty area.
- Front to front contact between robots with the ball between them does not constitute pushing.
- Any robot proceeding to the ball whose side (i.e. arm, shoulder etc.) makes contact with another robot cannot be called for pushing, even if the second robot is not proceeding to the ball.
- A robot pushed by another robot cannot simultaneously be called for pushing itself. Only the robot, who initially pushed the other robot, will be called for pushing.

4.11 Playing With Arms/Hands

Playing with arms/hands occurs when a field player or a goalkeeper outside its own penalty area moves its arms/hands to touch the ball (except during a fall or get-up). The goalkeeper is allowed to touch the ball with its arms/hands while it is within its own penalty area. A robot playing with arms/hands will be subject to the standard removal penalty and the ball will be replaced at the point where it contacted the arms/hands of the offending robot. If an own goal is scored as a result, the goal should count and the player should not be penalized.

Accidental playing with arms/hands when a robot falls or executes a get-up routine will not be penalized. If the ball goes out of play in this case, normal kick-in rules will apply (see Section 3.8). However, goals (except for own goals) resulting from a ball contact with the arms/hands during a fall or get-up do not count and result in a Goal Kick (see Section 3.9) as if the ball went over the goal line next to the goal.

4.12 Leaving the Field

A robot that intends to leave the 7.4 m × 10.4 m carpeted area will be subject to the standard removal penalty (see Section 4.2). The head referee will call “Leaving the Field <robot>”.

Additionally, a robot will also be subject to the standard removal penalty, when:

- The robot walks into the goalposts or goal net for more than 5 s, this includes robots that are stuck on the goalposts and unable to free themselves.
- The robot’s fingers become entangled in the net (without any time constraint).

- 静止的机器人因推搡而受罚是不合理的，包括那些在踢球时踢球的机器人，前提是球在踢球动作开始时足够近，踢球本可以成功。
- 正在站起的机器人不能因为推进而被处罚。
- 门将在看球或追球时，因推搡而受罚是不合理的，在其自己的罚球区内。
- 机器人之间有球时，相互接触不构成推搡。
- 任何向球移动的机器人，如果其侧面（例如手臂、肩膀等）与其他机器人接触，即使第二个机器人没有向球移动，也不应被指控推搡。
- 被另一个机器人推搡的机器人不能同时被指控自己推搡。只有最初推搡另一个机器人的那个机器人，才会被指控推搡。

4.11 手臂/手部触球

手臂/手部触球是指场内球员或在本方罚球区外的守门员移动手臂/手部触碰球（不包括摔倒或起身时）。守门员在本方罚球区内可以用手臂/手部触碰球。使用手臂/手部触球的机器人将受到标准罚球处罚，并且球将在犯规机器人手臂/手部接触点被替换。如果因此进球，进球有效，球员不会被处罚。

机器人摔倒或执行起身动作时意外触球不处罚。如果此时球出界，将适用正常界外球规则（见第3.8节）。然而，摔倒或起身动作期间球与手臂/手部接触导致的进球（不包括乌龙球）无效，并将执行界外球（见第3.9节），如同球越过球门线一样。

4.12 离开场地

打算离开 7.4m × 10.4 m 铺设的场地的机器人将受到标准移除处罚（参见第 4.2 节）。主裁判将喊出“离开场地 <机器人>”。

此外，当机器人出现以下情况时，也将受到标准移除处罚：

- 机器人进入球门或球网超过 5 秒，这包括卡在球门上且无法脱身的机器人。
- 机器人的手指在网中缠绕（没有时间限制）。

4.13 Jamming

During a match, any robot shall never jam the communication and the sensor systems of the opponents:

Wireless communication. Each robot is only allowed to send a limited number of UDP messages that have to comply with a predefined format (see Section 2.4.2). If a robot uses a different protocol or sends too much data in a game, penalties will apply. If a team violates this rule in multiple games, disqualification from the tournament (including all technical challenges and side competitions) as well as an entry in the penalty list will be the consequence. Except for the wireless cards and the access points provided by the organizers of the competition, nobody close to the field is allowed using 2.4 GHz or 5 GHz radio equipment (including cellular phones and/or Bluetooth devices).

Whistle interference. Both the teams and the audience shall avoid intentionally confusing the robots by producing similar sounds to the game whistle.

Acoustic communication. If acoustic communication is used by both teams, they shall negotiate before the match how they can reduce interference. If only one team uses acoustic communication, the robots of the other team shall avoid producing any sound. In addition, both the teams and the audience shall avoid intentionally confusing the robots by producing similar sounds to those used for communication.

Infrared communication. If infrared communication is used by both teams, they shall negotiate before the match how they can reduce interference (if at all). Both the teams and the audience shall avoid confusing the robots by producing similar infrared signals to those used for communication.

Visual perception. The use of flashlights is not allowed during the games. However, flash photography from the audience is allowable as long as the head referee believes the purpose of the flash is not to jam any of the robots.

4.14 Aborting A Penalty

If a penalty was incorrectly applied to a robot and acknowledged by the referee to be incorrect, the robot should first be removed and placed outside the field, following the same process for the Standard Removal Penalty (see Section 4.2). Once placed, the robot should then be allowed to continue play immediately after being placed and the GameController removes its penalty.

The aborted penalty will not be added to the total team penalty count and will not increase by 10 s, according to the standard removal penalty rules.

4.13 干扰

在比赛中，任何机器人不得干扰对手的通信和传感器系统：

无线通信。每个机器人仅被允许发送有限数量的UDP消息，且必须符合预定义的格式（参见第2.4.2节）。如果一个机器人使用不同的协议或在比赛中发送过多数据，将受到处罚。如果一个团队在多场比赛中违反此规则，后果将是被取消比赛资格（包括所有技术挑战和附属比赛）以及被列入处罚名单。除了由比赛组织者提供的无线网卡和接入点外，任何靠近赛场的个人均不得使用2.4GHz或5GHz无线电设备（包括手机和/或蓝牙设备）。

哨声干扰。双方团队和观众都应避免通过制造与比赛哨声相似的声响来故意干扰机器人。

声学通信。如果双方团队都使用声学通信，它们应在比赛前协商如何减少干扰。如果只有一方团队使用声学通信，另一方的机器人应避免发出任何声音。此外，双方团队和观众都应避免通过制造与通信时使用的相似声响来故意干扰机器人。

红外通信。如果双方队伍都使用红外通信，则应在比赛前协商如何减少干扰（如果需要的话）。双方队伍和观众都应避免通过产生与通信时使用的相似红外信号来混淆机器人。

视觉感知。比赛期间不允许使用手电筒。然而，只要裁判长认为闪光灯的目的不是干扰任何机器人，观众可以进行闪光拍照。

4.14 取消罚球

如果对机器人错误地应用了罚球，并且裁判确认该罚球是错误的，则应首先将机器人移出场地，并按照标准移除罚球程序（见第4.2节）进行处理。放置后，在GameController移除其罚球后，机器人应立即继续比赛。

根据标准移除罚则，被取消的任意球不会计入总团队罚球数，且不会增加10秒。

5 Judgment

The referees are the only persons permitted on the carpeted area (i. e. the field and the border area).

5.1 Head Referee

The head referee is in charge of the game. Any decision of the head referee is valid. The head referee's decision is final and can not be changed afterwards, even by video proof. There is no discussion about decisions during the game, neither between the assistant referees and the head referee, nor between the audience or the teams and the head referee.

The head referee announces decisions by a clear loud call, and (as required) whistle sound. The whistle, or where there is no whistle the first verbal word of the referees calls, defines the point in time at which the decision is made. The referees should make efforts to use consistent and clear calls, and it is preferable for referees to use the calls as specified in these rules.⁵ The intention of specifying the referee calls is for clarity and consistency across games.

Where a whistle is required, the head referee first whistles and then announces the reason for the whistle. The head referee may choose to use any normal sports whistle. Each whistle sound should be short and not too loud as to interfere with other fields and simultaneous games. The head referee must *only* sound the whistle in circumstances described in these rules. There are three circumstances when the whistle is sounded, kick-off (see Section 3.7), a goal (see Section 3.3), and ending a half of gameplay (see Section 3.1).

The head referee should avoid handling the ball (except for placing a ball for kick-off), and avoid handling the robots. Their duty is to monitor and adjudicate the game. The head referee should only handle robots and the ball if absolutely necessary to expedite gameplay or removal of penalized robots, where the assistant referees are otherwise occupied or too far away.

The head referee should be equipped with a suitable referee jersey, whistle, coin, and black or dark-blue socks.

The head referee may decide at any point before or during a game to relocate any objects around the field, or direct persons to another position around the field.

5.2 Assistant Referees

The two assistant referees handle the robots and the ball. They take the robots out when they are penalized, and they put the robots in again. If a team requests to pick up a robot, an assistant referee

⁵The calls specified in these rules are detailed in English. With the agreement of the teams, the referees may use suitable calls in any language. The exception to this are technical challenges that depend on the calls as specified.

5 判决

裁判员是唯一被允许在铺有地毯的区域（即球场和边界区域）的人员。

5.1 主裁判

主裁判负责比赛。主裁判的任何决定都有效。主裁判的决定是最终决定，之后即使有视频证据也不能更改。比赛过程中不得讨论决定，无论是助理裁判与主裁判之间，还是观众或队伍与主裁判之间。

主裁判通过清晰的响亮呼喊（并根据需要）哨声来宣布决定。哨声，或在没有哨声时裁判说的第一个词语，定义了决定做出的时间点。裁判应努力使用一致且清晰的呼喊，并且最好使用本规则中规定的呼喊。⁵ 规定裁判呼喊的目的是为了比赛的清晰性和一致性。

需要哨声时，主裁判首先吹哨，然后宣布哨声的原因。主裁判可以选择使用任何常规体育哨。每个哨声应简短且音量适中，以免干扰其他场地和同时进行的比赛。主裁判必须仅在规则所述情况下吹哨。吹哨有三种情况：开球（见第3.7节）、进球（见第3.3节）和结束半场比赛（见第3.1节）。

主裁判应避免触球（除开球外），并避免触碰机器人。他们的职责是监控和判罚比赛。主裁判只有在绝对必要时（例如为了加快比赛节奏或移除受罚机器人），且助理裁判另有任务或距离过远时，才可触碰机器人或球。

主裁判应配备合适的裁判服、哨子、硬币，以及黑色或深蓝色袜子。

主裁判可在比赛开始前或比赛过程中，决定将场内任何物品移至其他位置，或指示人员前往场内其他位置。

5.2 助理裁判

两名助理裁判负责处理机器人与球。他们会将受罚的机器人移出，并在需要时将其放回。若队伍要求捡起机器人，助理裁判

⁵这些规则中指定的呼叫以英语详细说明。经团队同意，裁判可以使用任何语言的适当呼叫。这一例外是技术挑战，它依赖于规则中指定的呼叫。

will pick it up and give it to one of the team members once the head referee approves. An assistant referee will also put the robot back on the field. An assistant referee will also replace the ball when it goes off the field or becomes stuck between a players feet.

The head referee should talk to the assistant referees before the game starts to determine what tasks or lesser decisions they wish to delegate to them, however the final calls must come from the head referee. Examples of delegated assistance to referee decisions include keeping track of indirect-kick rule compliance to help confirm if a valid goal has been scored and indicating possible rules violations requiring a penalty (such as leaving the field, or motion in set).

Assistant referees should only enter the field to execute a decision made by the main referee. They should not prevent robots from falling during the game.

The assistant referees should be equipped with a suitable referee jersey, and black or dark-blue socks.

5.3 GameController Operator

The operator of the GameController sits at a PC in the technical area. As with the head referee, the operator should make efforts to use consistent and clear calls. They will signal any change in the game state to the robots via the wireless as they are announced by the head referee. Note that for both kick-offs and goals, the moment of whistling is determining, not the verbal announcement of the head referee. The operator will also inform the assistant referees when a timed penalty is over and a robot has to be placed back on the field. They should announce events that occur automatically in the GameController due to elapsed time, such as the ball coming into play after a kick-off, penalty kick or free kick, or the state changing from ready to set. They are also responsible for keeping the time of each half. They should count aloud the remaining seconds in a half once the time remaining is 5 s or less. Finally, they should repeat the calls of the head referee to make sure it was heard correctly.

5.4 Referee-Team Communication

Both teams send a representative called team captain to the field 10 min before their match starts. This time should be used to welcome each other, assign team colors (see Section 2.2), choose side and kick-off (see Section 3.5), and discuss match related topics.

During the match only the team captains are allowed to communicate with the head referee. Only the team captains and two more people per team are allowed to stay next to the game controller tables. The rest of the team locates themselves around the other sides of the field if they want to watch the match. This allows the referees easier communication with the team and the game controller operator gets less disturbed.

一旦主裁判批准，将捡起它并交给团队的一名成员。助理裁判也会将机器人放回场上。当球出界或卡在球员脚间时，助理裁判也会更换球。

主裁判应在比赛开始前与助理裁判沟通，确定他们希望将哪些任务或次要判罚委托给他们，但最终判罚必须由主裁判做出。将裁判决策委托给助理裁判的示例包括：记录间接任意球规则的执行情况，以帮助确认是否已进球有效，以及指示可能需要判罚的规则违规行为（例如越位或动作犯规）。

助理裁判只能进入场地执行主裁判做出的判罚。他们不应阻止机器人在比赛中摔倒。

助理裁判应配备合适的裁判服，以及黑色或深蓝色的袜子。

5.3 控制器操作员

控制器操作员坐在技术区域的电脑前。与主裁判一样，操作员应努力使用一致且清晰的判罚。他们将根据主裁判宣布的游戏状态变化，通过无线方式向机器人发出信号。请注意，对于开球和进球，哨声的时机决定性，而不是主裁判的口头宣布。操作员还将通知助理裁判，当计时罚时结束且机器人需要被放回场地时。他们应宣布由于时间流逝而在控制器中自动发生的事件，例如开球、罚球或任意球后球进入比赛状态，或状态从准备变为就位。他们还负责记录每半场的时间。当剩余时间少于5秒时，他们应大声报出剩余秒数。最后，他们应重复主裁判的判罚，以确保被正确听到。

5.4 裁判与球队沟通

两支球队会在比赛开始前10分钟派一名被称为队长的人到场上。这段时间应用来互相欢迎、分配球队颜色（见第2.2节）、选择场地和开球（见第3.5节），以及讨论与比赛相关的话题。

比赛期间，仅允许队长与主裁判沟通。每队仅允许队长和另外两人靠近游戏控制器桌。其余队员若想观看比赛，需自行在场地其他区域就位。这便于裁判与队伍沟通，也让游戏控制器操作员不易被打扰。

After the match the teams thank the referees for their duty.

During all phases of the match teams and referees are communicating with respect to each other.

5.5 Referee List For Friendly Games

During the RoboCup competition setup days and during the game days, several teams may want to participate in friendly games with each other. In this case, there should be a list of all the head and assistant referees that are willing to judge a game. The referees on the list should be voluntary but participation should be encouraged. This is especially recommended for those who wish to gain referee experience. Teams are still free to choose their own referees.

The OC should be in charge of maintaining the list and should be approached if one should want to volunteer.

5.6 Referees During the Match

The head referee and the assistant referees should wear socks *of black or dark blue color* and avoid reserved colors (white and green) in their leg clothing. They may enter the field in particular situations, e. g., to remove a robot when applying a penalty. They should avoid interfering with the robots as much as possible.

5.6.1 Visual signal

For visual signal execution, the referee should have bare hands as much as possible. Exceptions apply as described in the following subsections and for small jewelry.

5.6.1.1 Challenge Shield The referee executing the signal will wear red gloves to increase visibility; however, teams are advised not to rely solely on this visual cue, as gloves may be omitted in future years.

比赛结束后，各队向裁判表示感谢。

在比赛的各个阶段，球队和裁判员之间应相互尊重地进行沟通。

5.5 赛前友好比赛裁判员名单

在RoboCup比赛设置日和比赛日期间，几支球队可能希望相互进行友好比赛。在这种情况下，应有一份所有愿意担任比赛裁判员的首席裁判和助理裁判的名单。名单上的裁判员应为自愿参与，但应鼓励他们参加。这尤其建议那些希望获得裁判经验的人。球队仍然可以自由选择自己的裁判员。

赛事组织者应负责维护该名单，如有意愿自愿参与，应联系他们。

5.6 比赛期间裁判员

首席裁判和助理裁判应穿着黑色或深蓝色颜色的袜子，并在腿部服装中避免使用保留颜色（白色和绿色）。在特定情况下，例如在执行罚球时移除机器人时，他们可以进入场地。他们应尽可能避免干扰机器人。

5.6.1 视觉信号

执行视觉信号时，裁判应尽可能保持双手裸露。以下小节所述情况及小件首饰除外。

5.6.1.1 挑战护盾 执行信号的裁判将佩戴红色手套以增加可见性；然而，建议各队不要仅依赖这一视觉提示，因为未来年份可能会取消手套。

A The Official RoboCup Competition Rules

This section contains rules that are not directly relevant for games and that may not apply at local opens. However, these rules will be upheld at the yearly international RoboCup competition.

A.1 Qualification Procedure and Code Usage

The qualification procedure as well as the corresponding deadlines will be announced by the Technical Committee before qualification applications are accepted.

The RoboCup Standard Platform League offers unique possibilities to use code from other teams. In spirit of the RoboCup every team is generally allowed to use code from other teams to push the league further with their own research. This use must be cited. However, every participant of RoboCup *has a duty* to contribute to the league.

To qualify, every team must make at least *one novel contribution* within their soccer software within the last 3 years. Contributions outside of this period are no longer considered sufficiently novel and a team must make at least one *new* contribution. It is also *mandatory* for a team to use their novel contribution in all competition games. A novel contribution is:

- Research publishable contribution to a *game critical module*
- Complete replacement of a *game critical module*, with original software. This may not necessarily be research publishable, but must be of equivalent scale and quality to research publishable work.

It is not a novel contribution to replace a module with code copied from another source, or to simply train a machine-learning model released by another team using new data.

As of the 2025 competition, the following are recognized as game critical modules: Ball detection, Robot detection, Robot vision (not otherwise listed), Localization, Walk/Kick engine, Dynamic stabilization, Behavior Architecture, & Distribution computation, Whistle detection.

As of the 2025 competition, the following are *not* recognized as sufficiently game critical (even if the ability to play soccer depends on these): Hand-written Soccer Behaviors, Natural Language detection, & Robot and GC Communication.

In their qualification application, teams may petition the technical committee to recognize other novel contributions not listed here. Additionally, a team that has participated at RoboCup for at least 3 years consecutively may petition the technical committee to recognize contributions to non-game critical modules, such as developing infrastructure for the league.⁶ A team may also

⁶However, the technical committee should balance whether a team is continuing to use their own software in games.

A 官方机器人世界杯竞赛规则

本节包含与游戏不直接相关的规则，且可能在地方赛中不适用。然而，这些规则将在每年的国际RoboCup竞赛中得到遵守。

A.1 资格审查程序和代码使用

资格审查程序及相应的截止日期将在资格审查申请被接受前由技术委员会公布。

机器人世界杯标准平台联赛为使用其他队伍的代码提供了独特的机会。本着机器人世界杯的精神，每个队伍通常都允许使用其他队伍的代码，以通过自己的研究推动联赛发展。这种使用必须注明出处。然而，机器人世界杯的每个参与者都有义务为联赛做出贡献。

要获得资格，每个队伍必须在过去3年内至少在其足球软件中做出一项新颖的贡献。此期间外的贡献不再被视为足够新颖，队伍必须至少做出一项新贡献。同时，队伍必须在所有比赛游戏中使用其新颖的贡献。新颖的贡献是：

- 对游戏关键模块的可发表研究成果
- 完全替换游戏关键模块，使用原始软件。这可能不一定能发表研究成果，但必须达到与可发表研究成果相当规模和质量的水平。

用从其他来源复制的代码替换模块，或仅使用新数据训练其他团队发布的机器学习模型，并非一项新颖的贡献。

截至2025年比赛，以下被认定为游戏关键模块：球体检测、机器人检测、机器人视觉（未另行列出）、定位、行走/踢球引擎、动态稳定、行为架构、&分布计算、哨声检测。

截至2025年比赛，以下不被认定为足够关键（即使踢球能力依赖于这些）：手写足球行为、自然语言检测、&机器人和GC通信。

在其资格申请中，团队可以向技术委员会申请认可此处未列出的其他创新贡献。此外，连续参加RoboCup至少3年的团队可以向技术委员会申请认可对非游戏关键模块的贡献，例如为联赛开发基础设施。⁶ 团队还可以

⁶然而，技术委员会应权衡团队是否继续在游戏中使用自己的软件。

petition for the technical committee to reconsider the list of game critical and non-game critical modules. Successful petitions will be publicly announced to the league for transparency.

If a team that is otherwise eligible for qualification cannot provide sufficient evidence of the required contributions by the deadline for applications, then that team may be qualified for RoboCup *on probation*. In this case, the team must provide evidence of the required contributions to become *fully* qualified by the registration deadline of the RoboCup event. If no suitable evidence is provided, the team's probationary qualification will lapse.

Every applicant must also bring a poster containing the team's contribution, focused on the current year, to the RoboCup event to share their contributions with the other teams.

Failure to meet any of these requirements will result in a qualification penalty for subsequent years.

A.2 Champions Cup and Challenge Shield

Two separate competitions are played: the Champions Cup for the strongest teams and the Challenge Shield for all other teams. Teams are assigned to one of the competitions during the application phase, considering their preference, ability and organizational aspects. Matches in the Champions Cup are played with up to 7 players on each team, while the Challenge Shield is played with up to 5 players on each team (see Section 2).

In addition, some rules are simplified for Challenge Shield matches. The major differences are:

- The requirement to recognize visual referee signals is relaxed for free kicks (see Section 3.9) as well by using red gloves (see Section 5.6.1).
- Although the indirect kick rules are the same for both competitions, Challenge Shield teams have the option to have them relaxed if they declare they cannot play with more than two robots. (See Section 3.4.1)

A.3 Game Structure

The clock stops during stoppages of play (such as `ready` and `set` state after goals) from the quarter-finals onward. In round robin pool play, a game can finish in a draw as no penalty shoot-out will follow. In the promotion round, intermediate round, quarter finals, semi finals, 3rd place or final, a game that ends in a draw will be followed by a penalty shoot-out (see Section 3.19).

申请技术委员会重新考虑游戏关键和非游戏关键模块的列表。成功的申请将向联赛公开宣布，以保持透明度。

如果一个团队在资格方面符合条件，但在申请截止日期前无法提供足够的证据证明其所需贡献，那么该团队可能被有条件地录取参加RoboCup。在这种情况下，该团队必须在RoboCup比赛的注册截止日期前提供所需贡献的证据，才能完全获得资格。如果未提供合适的证据，该团队的有条件资格将失效。

每位申请人都必须携带一份海报，展示团队当年的贡献，带到RoboCup活动现场，与其他团队分享其贡献。

未能满足任何一项要求，将在后续年份导致资格处罚。

A.2 拔河杯和挑战盾

设有两个独立的比赛：最强队伍的拔河杯和所有其他队伍的挑战盾。在申请阶段，根据队伍的偏好、能力和组织因素，将队伍分配到其中一个比赛。拔河杯的比赛每队最多7名选手，而挑战盾的比赛每队最多5名选手（见第2节）。

此外，挑战盾赛中的一些规则有所简化。主要区别是：

- 对于任意球，识别视觉裁判信号的要求有所放宽（见第3.9节）以及使用红色手套（见第5.6.1节）。
- 虽然两种比赛的间接球规则相同，但挑战盾赛队伍可以选择放宽规则，如果他们声明无法使用超过两台机器人进行比赛。（见第3.4.1节）

A.3 游戏结构

比赛暂停期间（例如进球后的准备和就位状态）从四分之一决赛开始，时钟会停止。在循环赛小组赛中，比赛可以打成平局，因为没有随后的点球大战。在升级赛、中间赛、四分之一决赛、半决赛、第三名决赛或决赛中，如果比赛打成平局，将进行点球大战（见第3.19节）。

A.4 Competition Mode

Depending on the number of participating teams, the OC and TC create a competition table. The following modes can be selected:

- Group games, with or without intermediate rounds, with concluding knock-out phase
- Double elimination system
- Swiss tournament system
- Swiss tournament system with concluding knock-out phase.

All teams participating in the competition are initially ranked using the Glicko system⁷ based on all available results from previous official RoboCup tournaments. (New teams will be ranked equally below all previously competing teams. Teams that participated previously but did not participate in the previous year will be ranked above new teams but below teams that competed in the previous year.)

The OC will announce two weeks before a competition:

- Competition mode and which competition rules will be applied
- Initial ranking of participating teams
- How to determine the overall winner and the ranking of the technical challenges
- Pre-qualification rules.

A.5 Referee Selection and Requirements

During pool play, the games will be refereed by members of teams from a different pool.

Each team has to referee a number of games. A schedule will be released specifying the games for which each team is required to provide two referees. Referees should report to the appropriate field at least ten minutes before the game is scheduled to start.

If a team fails to provide two referees for a game in which they are scheduled to provide referees, it will be noted by the organizing committee and recorded as a **qualification penalty** (Appendix A.9).

For each of the games, a team will be required either to provide the head referee and the operator of the GameController, or the two assistant referees. The two teams assigned to referee a game shall

⁷<http://www.glicko.net/glicko/glicko.pdf>

A.4 比赛模式

根据参赛队伍的数量，OC和TC会创建一个比赛表。可以选择以下模式：

- 小组赛，可包含或不包含中间轮次，最终以淘汰赛形式决出胜负
- 双败淘汰制
- 瑞士制赛制
- 瑞士制赛制，最终以淘汰赛形式决出胜负

所有参加本次比赛队伍的初始排名均使用Glicko系统⁷，基于所有先前官方RoboCup赛事的可用成绩进行计算。（新队伍将排名在所有先前参赛队伍之下。先前曾参赛但未参加去年的队伍将排名在新队伍之上，但在去年参赛队伍之下。）

OC将在比赛前两周公布：

- 比赛模式和将适用的比赛规则
- 参赛队伍的初始排名
- 如何确定总胜者和技术挑战的排名
- 预选规则。

A.5 裁判选拔与要求

小组赛阶段，比赛将由不同小组的球队成员担任裁判。

每支球队都需要担任一定数量的比赛裁判。将发布赛程表，明确每支球队需要提供两名裁判的比赛。裁判应在比赛预定开始前至少十分钟到达指定场地。

如果一支球队未能为需要其提供裁判的比赛提供两名裁判，组织委员会将予以记录，并记为资格处罚（附录A.9）。

对于每场比赛，一个队伍需要提供主裁判和游戏控制器的操作员，或者两名助理裁判。被分配负责一场比赛的两队应

⁷<http://www.glicko.net/glicko/glicko.pdf>

decide among themselves which roles each team will fulfill. Note, however, that the head referee and the GameController should always be from the same team.

A team may swap their scheduled refereeing duties with another team, but the team listed on the referee schedule will be held accountable if referees fail to appear for a game they are scheduled to referee.

The requirement to referee may be an extreme hardship for extremely small teams. If a team believes providing two referees for games will be an extreme hardship, they must send an email explaining their situation to the Organizing Committee and Technical Committee at least two weeks before the first set up day of the competition. The Organizing and Technical Committees will then consider the request and attempt to find an acceptable solution.

Referees must have good knowledge of the rules as applied in the tournament, and the operator of the GameController must be experienced in using that software. Referees and the GameController should be selected among the more senior members of a team, and preferably have prior experience with games in the RoboCup Standard Platform league.

In each game, each of the teams playing shall be able to veto one and only one eligible referee with no reason required. The veto must be delivered before the start of the `standby` phase or during a timeout.

A.5.1 Best Referee Voting

Teams will be asked to vote for the team who has provided the best referees throughout the competition. Specifics on the voting process and awards will be detailed by the Organizing Committee during a referee meeting before the start of the competition.

A.6 Rules for Forfeiting

Teams who do not make a good faith effort to participate in a scheduled game are considered to forfeit the game.

If a team notifies the technical committee that they wish to forfeit less than two hours before their scheduled game time, simply fails to show up for their game, or decides during their game that they wish to forfeit, then the opposing team will play the match against an empty field. However, any own goals will not be scored. Hence, after an opponent forfeits, the team playing against an empty field cannot do worse than they were doing at the time the opponent decided to forfeit. Teams may choose to forfeit at any stoppage of play. However, once a forfeit is announced, they may not reverse this decision.

If a team notifies the technical committee that they wish to forfeit at least two hours before their schedule game time, the following procedure will be followed.

自行决定每队将承担哪些角色。然而，需要注意的是，主裁判和游戏控制器操作员必须始终来自同一队。

一个团队可以与另一个团队交换他们的赛程裁判职责，但如果裁判未能出席他们被安排裁判的比赛，则列在裁判安排表上的团队将承担责任。

担任裁判的要求可能对极小的团队来说是极大的困难。如果一个团队认为为比赛提供两名裁判将构成极大的困难，他们必须在比赛第一轮设置日前至少两周前，向组委会和技术委员会发送电子邮件解释他们的情况。然后，组委会和技术委员会将考虑该请求并尝试找到一个可接受的解决方案。

裁判必须熟悉比赛规则，并且GameController的操作者必须具备使用该软件的经验。裁判和GameController应从团队中更资深的成员中选拔，最好有RoboCup标准平台联赛的赛事经验。

在每场比赛中，每支参赛队伍都有权否决一名符合条件的裁判，无需提供理由。否决必须在待命阶段开始前或暂停期间提出。

A.5.1 最佳裁判投票

将要求各队投票选出在本次比赛中提供最佳裁判服务的队伍。关于投票流程和奖项的具体细节，组委会将在比赛开始前举行的裁判会议上进行详细说明。

A.6 弃权规则

未以诚意参与预定比赛且未做出合理努力的队伍将被视为弃权。

如果队伍在预定比赛开始前不到两小时通知技术委员会他们希望弃权、无故未出席比赛，或决定在比赛过程中弃权，那么对方队伍将空场进行比赛。然而，任何自摆进球都不会被算作有效进球。因此，在对方弃权后，空场作战的队伍表现不会比对方决定弃权时更差。队伍可以在任何比赛停顿时选择弃权。但是，一旦弃权决定被宣布，他们便不能撤销该决定。

如果一个团队在比赛时间至少提前两小时通知技术委员会他们希望弃权，则将遵循以下程序。

- If a team chooses to forfeit a match in the round robin games the other team plays the match against an empty field. However, any own goals will not be scored.
- If a team chooses to forfeit in a knock-out game it gets replaced by the next best qualified team, i. e. the team it kicked out or left behind in the round robins.

Note that there are a few unlikely cases that are not covered by these rules. If a situation is not covered by these rules, the technical committee and the organizing committee will work together to make a decision.

Any forfeit will result in a qualification penalty being recorded (see Appendix A.9) but the circumstances of the forfeit will affect the severity of the offense and the impact on future qualification.

A.7 Source Code Releases

All teams that have participated in RoboCup must subsequently release code from that year’s codebase. The code must be licensed such that other RoboCup participants can use it, although the license may place conditions on its use. The preferred type of release is the full source code of the software that was running in the team’s last game at RoboCup. In case this is not possible (e. g. due to legal reasons), it is required that at least the source code related to the novel contributions (as given during the qualification process) is published. Participation in technical challenges may come with additional requirements on the amount of components to be released.

The source code must be published and its availability announced on the SPL mailing list (`robocup-nao@lists.robocup.org`) by 2025-10-15. Failing to publish source code by the deadline will result in a qualification penalty being recorded (see Appendix A.9).

A.8 Subsequent Year Pre-qualification Procedure

Teams may become pre-qualified for the subsequent year’s competition by fulfilling criteria set two weeks before the competition. Pre-qualified teams do not compete with other teams’ applications during the qualification process. However, the call for participation for the subsequent year’s competition may set additional formal requirements that must be fulfilled in order to remain pre-qualified.

A.9 Qualification Penalties

There are a number of offenses which lead to qualification penalties being recorded against a team. These are as follows:

- 如果一个队伍在循环赛中选择弃权比赛，那么另一支队伍将空场进行比赛。但是，任何进球都不会被计算。
- 如果一个团队在淘汰赛中选择弃权，则会被下一支最符合条件的队伍替换。队伍，即被淘汰或在循环赛中留下的队伍。

请注意，有一些不太可能的情况这些规则没有涵盖。如果一个情况没有被这些规则涵盖，技术委员会和组织委员会将共同做出决定。

任何弃权都会记录资格处罚（见附录A.9），但弃权的具体情况会影响违规的严重程度以及对未来资格的影响。

A.7 源代码发布

所有参加RoboCup的队伍都必须随后发布该年代码库的代码。代码必须获得许可，以便其他RoboCup参与者可以使用它，尽管许可可能会对其使用提出条件。首选的发布类型是团队在RoboCup最后一场比赛中运行的软件的完整源代码。如果无法做到这一点（例如由于法律原因），则要求至少发布与新颖贡献（如在资格过程中提供的）相关的源代码。参加技术挑战可能会对要发布的组件数量提出额外要求。

源代码必须在2025年10月15日前发布，并在SPL邮件列表（`robocup-nao@lists.robocup.org`）上公布其可用性。未能按期发布源代码将导致记录资格处罚（参见附录A.9）。

A.8 下一年度预资格程序

队伍可以通过在比赛前两周满足设定的标准来获得下一年度比赛的预资格。获得预资格的队伍在资格过程中不与其他队伍的申请竞争。然而，下一年度比赛的参与邀请可能会设定额外的正式要求，必须满足这些要求才能保持预资格。

A.9 资格处罚

有一些违规行为会导致记录在队伍上的资格处罚。具体如下：

- Withdrawing from RoboCup after the final commitment deadline
- Failing to referee when assigned (Appendix A.5)
- Forfeiting a game (Appendix A.6)
- Failing to publish source code by the deadline (Appendix A.7)

A team cannot be pre-qualified for RoboCup in the year following a qualification penalty. Furthermore, a qualification penalty is considered by the Technical Committee when reviewing applications and will negatively affect the assessment of a team’s application. Multiple penalties accumulate and will result in an even more negative assessment of a team’s application. Qualification penalties are considered for a period of three years following the offense.

Whenever a qualification penalty is recorded, all relevant details including any possible mitigating circumstances are also recorded and these will also inform the assessment of a team’s application.

A.10 Disqualification During Competition

A team may be disqualified during the RoboCup competition for:

- A serious violation of the terms of a team’s qualification
- Gaining a Qualification Penalty during the course of the competition (see Appendix A.9)
- A serious breach of ethics, or serious behavior unbecoming of participants of RoboCup.

Example. A team promises to use their novel contribution in RoboCup games, but fails to do so. Alternatively, a team deliberately misleads the technical committee about the novelty of their work and/or their contribution to the league, such that they are deemed to have copied another team.

A team can *only* be disqualified by a decision of the *Board of Trustees of the RoboCup Federation*. The RoboCup Soccer SPL executive must petition the board in writing at their soonest possible availability. The executive must simultaneously inform the relevant team of the petition in writing.

A disqualified team automatically forfeits all games (see Appendix A.6). For practicality, the disqualification should not apply *retroactively*. However, by majority vote of the team leaders, provisions for retroactive disqualification may be made in the fairness of the affected teams.

- 在最终承诺截止日期后退出RoboCup
- 未在指定时担任裁判（附录A.5）
- 弃权比赛（附录A.6）
- 未能在截止日期前发布源代码（附录A.7）

一个队伍在受到资格处罚的次年不能被预选为RoboCup。此外，资格处罚将在技术委员会审核申请时被考虑，并会对队伍的申请评估产生负面影响。多次处罚会累积，导致对队伍申请的评估更加负面。资格处罚将在违规行为发生后三年内被考虑。

每当记录资格处罚时，所有相关细节包括任何可能的减轻情况也会被记录，这些信息也将用于评估队伍的申请。

A.10 比赛期间取消资格

一个团队可能在RoboCup比赛中被取消资格，原因包括：

- 严重违反了团队资格条款
- 在比赛过程中获得资格处罚（参见附录A.9）
- 严重违反道德，或行为有失RoboCup参赛者身份。

示例。一个团队承诺在RoboCup比赛中使用其创新贡献，但未能兑现。或者，一个团队故意误导技术委员会关于其工作的创新性以及/或其对联赛的贡献，以至于被认为抄袭了另一支团队。

一个团队只能被RoboCup联合会受托人委员会的决定取消资格。RoboCup足球SPL执行委员会必须在最短的时间内以书面形式向委员会提出申请。执行委员会必须同时以书面形式通知相关团队该申请。

一个被取消资格的队伍将自动放弃所有比赛（参见附录A.6）。出于实用性考虑，取消资格不应追溯适用。但是，经队伍负责人多数投票通过，可以出于对受影响队伍的公平性考虑，制定追溯性取消资格的规定。

B Changes From 2024

This is a brief non-normative list of rule changes from 2024 to 2025:

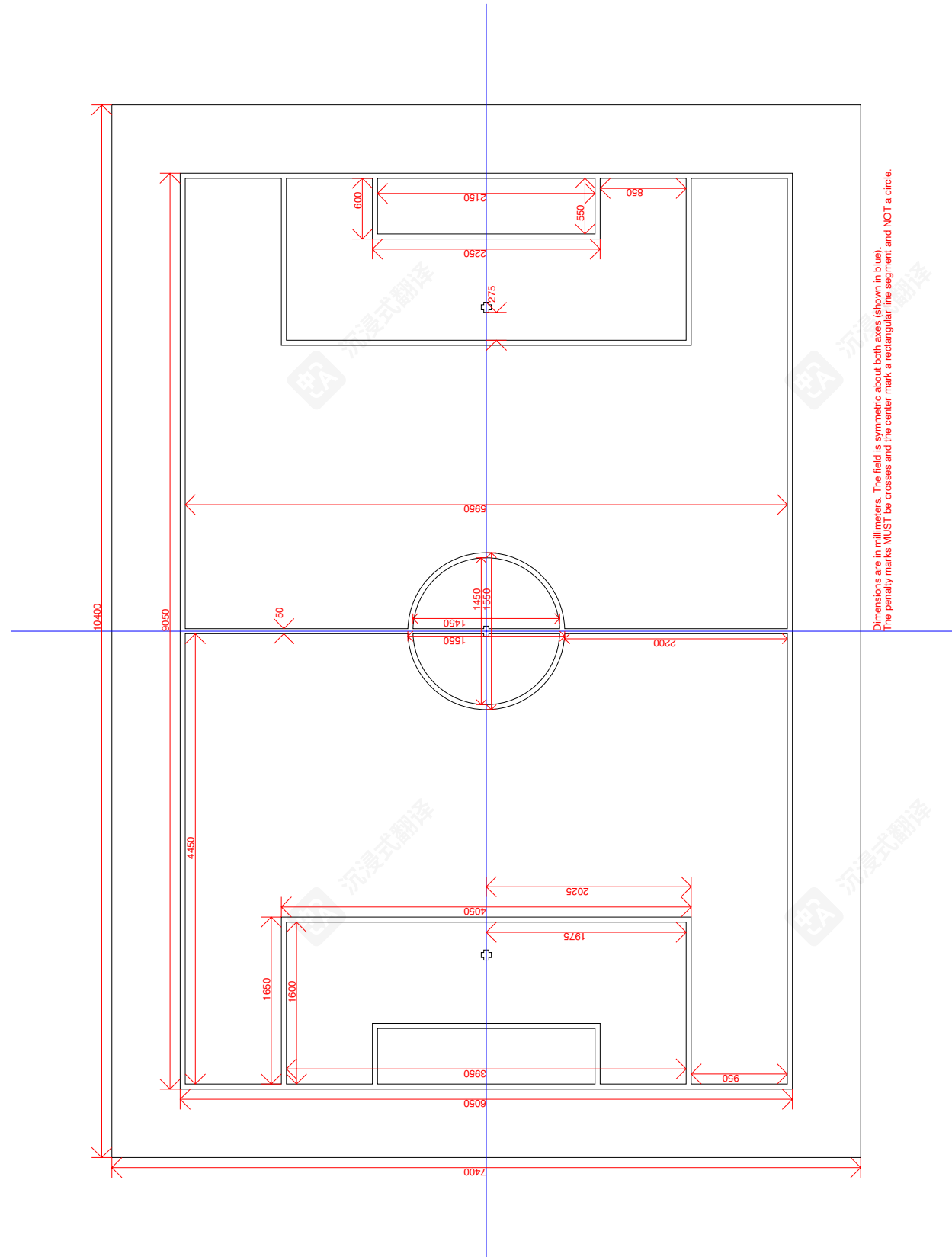
- The `ready` visual signal and the corresponding delay also apply to Challenge Shield as well as Champions Cup (see Section 3.2).
- The `ready` signal delay is increased to 45 s for Champions Cup and to 35 s for Challenge Shield (see Section 3.2).
- The `Motion in Standby` penalty for falsely detecting the `ready` visual signal now results in a team wide penalty, but for a shorter duration (see Section 4.5.1).
- Team communication packets sent during `standby` now count down the team’s message budget for the game (see Section 2.4.2).
- The kicking team for free kicks in Champions Cup matches is communicated via visual referee signal. The `GameController` no longer shares this information in those games (see Section 3.9.1).
- If a team fails to execute a free kick, the opposing team may attempt to score directly (see Section 3.4).
- The indirect kick rules from Champions Cup 2024 now applies to Challenge Shield as well. Challenge Shield teams have a fallback option available, if they are unable to field enough robots (see Section 3.4).
- In special situations where it can’t be determined which team should have kickoff, neither team gets to play the ball first, instead the drop ball rule applies (see Section 3.17).
- Penalties can now be aborted early in case they were erroneously assigned by the referee (see Section 4.14).
- With the ball stop rule each half of the game can be allowed to last a little longer to let the attack situation finish (see Section 3.18).

B 2024年变更

这是从2024年到2025年的简要非规范性规则变更列表：

- 准备就绪的视觉信号及其对应的延迟同样适用于挑战护盾和冠军杯（参见第3.2节）。
- 冠军杯的准备信号延迟增加到45秒，挑战护盾则为35秒（参见第3.2节）。（参见第3.2节）。
- 在待机状态下错误检测准备视觉信号现在会导致全队受罚，但持续时间更短（参见第4.5.1节）。{v1}但持续时间更短（参见第4.5.1节）。
- 备用期间发送的团队通讯包现在会倒计时团队的剩余消息游戏预算（见第2.4.2节）。
- 在欧冠比赛中，角球的开球团队通过裁判的视觉信号进行沟通。游戏控制器不再在这些比赛中共享此信息（见第3.9.1节）。
- 如果一支球队未能执行角球，对方球队可以尝试直接得分（见第3.4节）。
- 欧冠2024年的间接球规则现在同样适用于挑战盾。如果挑战盾队伍无法部署足够的机器人，他们可以选择备用方案（见第3.4节）。
- 在特殊情况下，如果无法确定哪个队伍应该获得开球权，则队伍先触球，而不是适用掷球入界规则（见第3.17节）。
- 判罚现在可以在裁判错误分配的情况下提前取消（参见第4.14节）。
- 通过球停规则，每半场可以允许稍微长一点的时间，以让进攻情况完成（参见第3.18节）。

C Field Technical Drawings



C 技术图纸

